

Photoshop worklist

54 pages measured · ordered easiest fix first · 11 August 2026

Every correction I have measured, with the file, the pixel to erase, the pixel to draw to, and what is under each. Grouped by **how hard the Photoshop job is**, not by how bad the error is — so the top of this document is where to start.

Read the two shortcuts in section 2 before you open anything. On about a third of these pages the correct fix is to move the *label text*, not the leader line — which is a clean cut-and-paste on flat background instead of repainting over illustration. That converts the hardest class of defect into the easiest.

1 Before you open anything

Decide who owns each page — you or production

These are **flat JPEGs**. Production's fix is to re-render the page from their source scene, which keeps every other pixel intact. Your fix is to repaint the shipped JPEG. **If production later re-renders a page you have edited, your work is silently overwritten.** So pick a lane per page and tell them: either "we've fixed these ourselves, skip them" or "leave these to you". Doing both is the one outcome that wastes the effort.

My suggestion: **take groups A and B yourself** (short, self-contained, and they are the ones production is most likely to get wrong a second time), and **leave group C and everything in group D to them.**

Where the files are

All paths are relative to the root of the **private app repo** (rounds-codex-app). Most galleries live under `assets/<id>/<id>-02.jpg`; three of the pages here sit at the repo root instead, and those three are also the ones that ship undersized:

GALLERY	FILE	PAGE SIZE
most galleries	<code>assets/<id>/<id>-02.jpg</code>	1024 × 1536
tb	<code>tb-02.jpg</code>	800 × 1200
pneumothorax	<code>pneumothorax-02.jpg</code>	800 × 1200
hypothyroid	<code>hypothyroid-02.jpg</code>	800 × 1200

Every coordinate in this document is in the page's own pixels, origin top-left, which is what Photoshop's Info panel shows. For the three 800×1200 pages, multiply by 1.28 if you are working from a re-exported 1024×1536 file.

Three things to do after every edit

#	WHAT	WHY
1	Save as JPEG, quality 10-12 , same filename, same dimensions	The app and the service worker key off the path. Do not rename, do not resize.
2	Regenerate the thumbnail — <code>gthumbs/<id>-02.jpg</code> , 320 px wide, quality 82	The gallery grid reads the thumb, not the page. An unregenerated thumb shows the old artwork on the card.
3	Rebuild the gallery PDF — <code>scripts/rebuild_gallery_pdf.py <root> <id></code>	This is the one that has already bitten us. The download PDF is a build artifact generated once. Four galleries shipped for months with the old logo in the PDF and the new one in the viewer, because nothing regenerated the PDF. Send me the edited pages and I will run steps 2 and 3.

Then `python3 scripts/verify_gallery_pdfs.py <root>` checks every page of every gallery PDF against the page the app serves — it will catch it if step 3 was missed.

2

Two shortcuts that save most of the work

Shortcut 1 — swap the label text, not the leader

Where two labels have landed on *each other's* structure, both leaders are already correct in the sense that each ends on a real structure that the *other* label names. So the fix is to **exchange the two pieces of label text** and leave every line untouched.

Label text sits on flat dark panel background. Lasso, cut, paste, nudge into place, heal the two vacated rectangles. **No painting over illustration at all.**

PAGE	SWAP THESE TWO LABELS	NOTE
cellulitis p2	Artery ↔ Vein	Single words, stacked one above the other in the right-hand column. The cleanest one on this list — do it first.
ckd p2	Bowman's capsule ↔ Glomerulus	RENAL CORPUSCLE panel. A pure swap.
cdiff p2	Normal mucus layer ↔ Intact epithelium	NORMAL COLONIC MUCOSA panel. A pure swap.
osteoporosis p2	Tibia ↔ Fibula	Single words, stacked. But see group 5 — confirm which bone is which on the lateral skeleton first, and note that on the <i>anterior</i> skeleton both dots are on one bone, so a text swap alone will not finish that half.

Shortcut 2 — shift a whole label column by one position

Three pages have a column of labels where **every label has taken the next one's structure**. Rather than moving several leaders, move the text down (or up) one slot in the column and fix only the one label at the end of the chain that has nowhere to go.

PAGE	THE CHAIN	WHAT IS LEFT OVER
labor p2	Sacrum → a lumbar vertebra · Rectum → the sacrum · Vagina → the rectum	Shift the three right-hand labels up one slot, then give the vagina its own leader. Erase the topmost dot at (732,341).
withdrawal p2	HYPOTHALAMUS → pink · AMYGDALA → green · LOCUS COERULEUS → orange · CEREBELLUM → orange · BRAINSTEM → nothing	Shifting the text fixes three of five. Two leaders still need moving: one onto the blue blob at (437,495) and one onto the mauve cerebellum at (505,530) , neither of which currently has any leader.
cdiff p2	Sigmoid colon → the descending limb · Rectum → haustrated bowel · Anal canal → sigmoid	Shift the three down one slot. Descending colon at (462,348) is correct and stays.

One caution on both shortcuts. Check the label's own ink colour before you move it. On several of these pages the label is printed in the colour of the structure it names — *withdrawal* p2 is entirely built that way — so moving the text without recolouring it would leave the colour wrong even though the words are right.

3

The erase-and-repaint method

Where you do have to move a line, this is what I learned doing about a dozen of these with scripts. Photoshop's healing brush will do most of it better than my code did, but the choice of *donor* is the whole job.

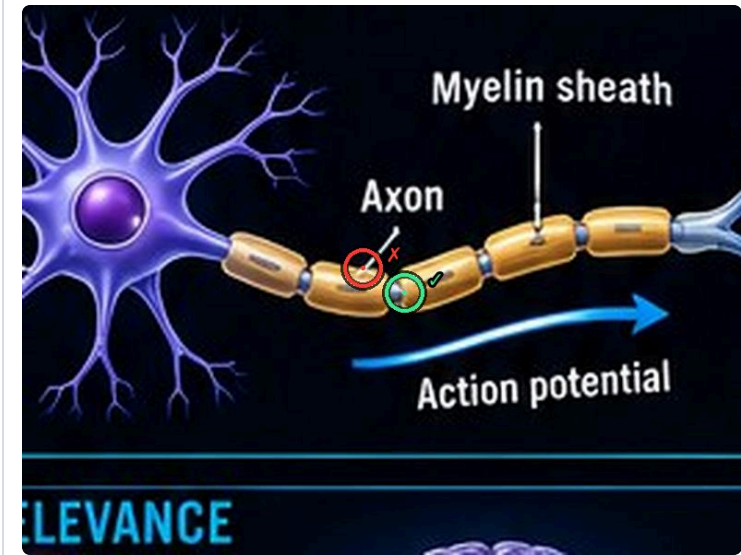
WHAT IS UNDER THE LINE	HOW TO ERASE IT
Flat black background	Anything works. Spot healing brush, one pass.
Flat or smoothly shaded tissue	Healing brush with a donor 20–30 px away, perpendicular to the stroke.
Textured tissue (gyri, trabeculae, folia)	Clone stamp, donor parallel to the stroke so the texture direction survives. Sample often; a single donor smears.
Banded or striated tissue (adrenal cortex, muscle, arbor vitae)	Clone along the band , not across it. Find the grain by eye first — on <i>addisons</i> p2 the leaders ran <i>parallel</i> to the striations, so the donor had to be perpendicular. The rule is "clone along the texture's grain", and the grain sometimes runs with the stroke.
A strong edge crosses the stroke — an organ boundary, a vessel wall, a bone cortex	Do not erase it. Any donor reconstructs one side and destroys the other. I abandoned two fixes for exactly this. Leave those for a re-render.

Erase the whole polyline, not the straight line between its ends. Several of these leaders bend. On *pancreatitis* p2 I masked a straight line between the endpoints and left a white diagonal across the aorta — the bounding box and the pixel count both looked right, and only a side-by-side caught it.

Check your work at 100%, then at 25%. A repaint that is invisible at 100% often shows as a smudge at thumbnail size, which is where most readers see the page first.

A **Group A — move one dot, easy background**

Start here. Each is a single terminator on flat or dark ground, moving under about 40 px. The red ring is what to erase; the green ring is where to put it.



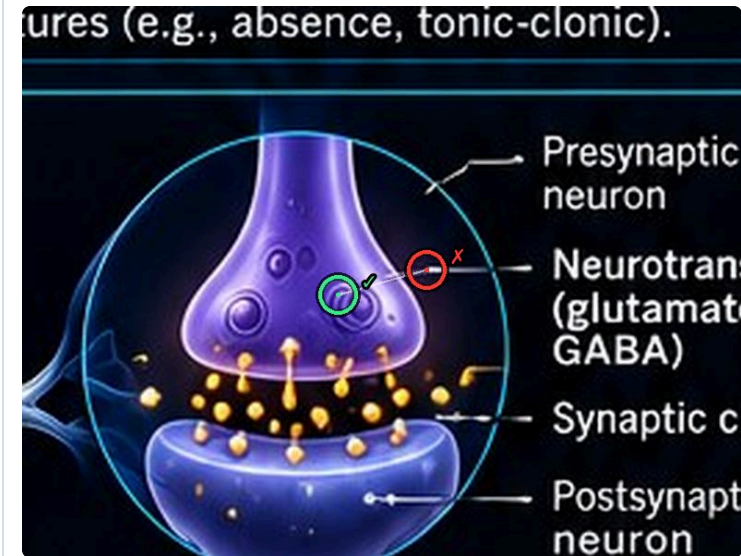
1

seizure p2 — Axon

x ERASE (253,1143)
a tan myelin internode — the same kind of segment the “Myelin sheath” label points at

✓ DRAW AT (268,1152)
a node of Ranvier, the blue-grey constriction between internodes

Nodes also at (300,1145) and (337,1130) if a different one suits the leader better.



2

seizure p2 — Neurotransmitters (glutamate, GABA)

x ERASE (527,1080)
the presynaptic bouton’s rim — no vesicle or particle within 25 px

✓ DRAW AT (495,1089)
a vesicle circle inside the bouton

Other vesicles at (473,1074) and (459,1092). Prefer a vesicle over the cleft — “Synaptic cleft” already points into the particle field.



3

depression p2 — Nucleus accumbens (Dopamine)

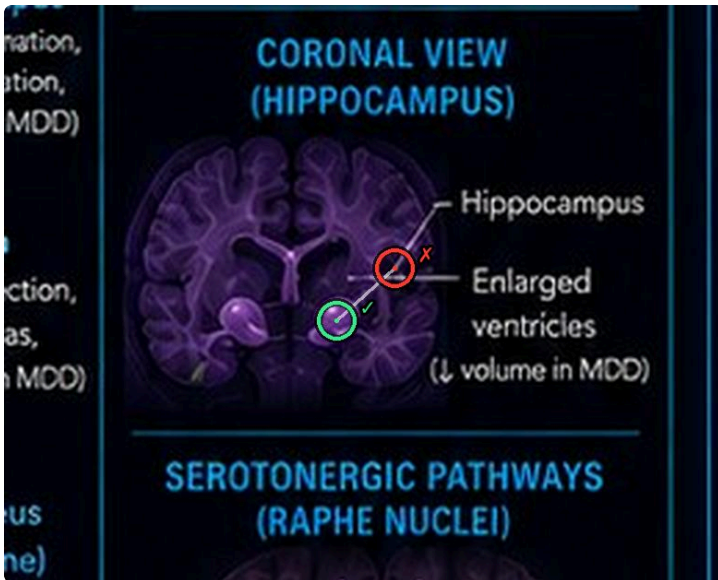
✗ ERASE (493,424)

the dark gap just below the sphere

✓ DRAW AT (492,401)

the blue-violet sphere, 23 px higher

Old dot is on near-black; erase is a one-pass heal.



4

depression p2 — Hippocampus (CORONAL VIEW)

✗ ERASE (905,415)

dark cortex / white matter

✓ DRAW AT (884,434)

the bright purple hippocampus

The only bright object in that neighbourhood.



5

stroke p1 — Thrombus (clot) blocking the MCA

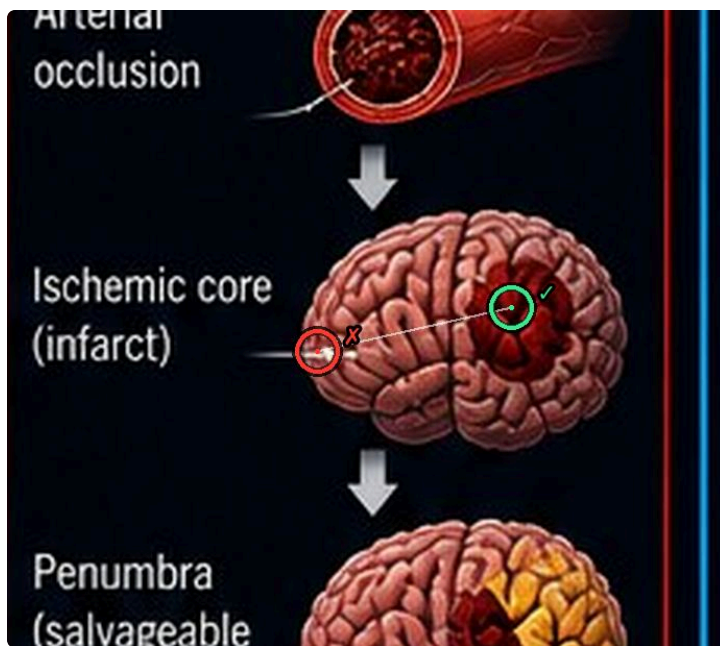
✗ ERASE (556,368)

inside the dark red infarct core — brain tissue

✓ DRAW AT (548,375)

the artery, 8 px away

The magnifier inset shows a clot inside a vessel lumen, so the anchor belongs on the vessel.



6

stroke p1 — Ischemic core (infarct)

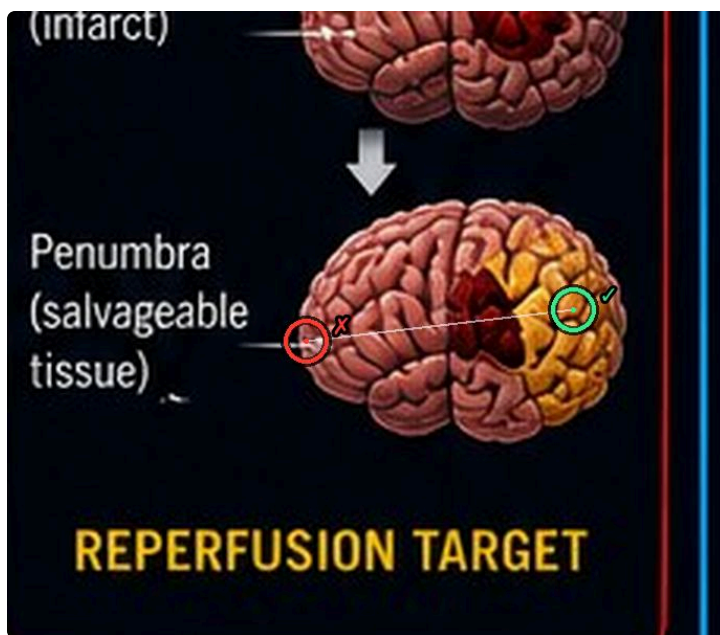
☒ ERASE (893,631)

the plain pink LEFT hemisphere of the icon

☒ DRAW AT (955,617)

the dark red core on the right hemisphere

The icon has exactly one coloured region — no ambiguity.



7

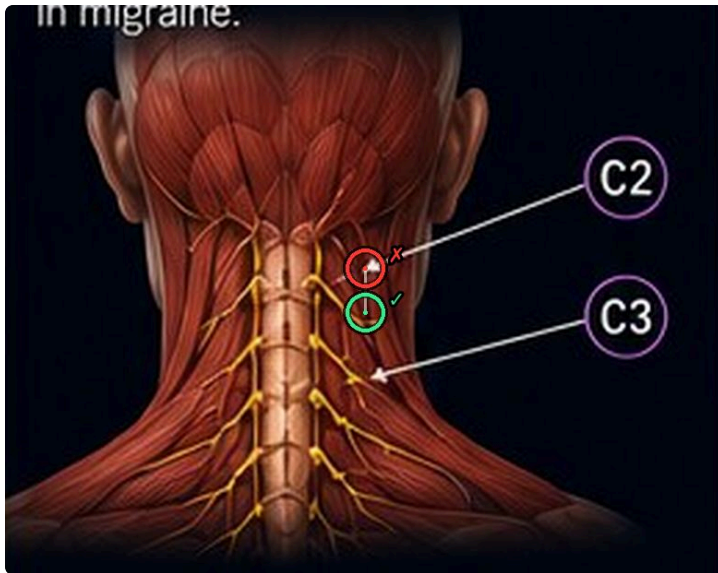
stroke p1 — Penumbra (salvageable tissue)

☒ ERASE (890,730)

the plain pink LEFT hemisphere

☒ DRAW AT (975,720)

the gold rim on the opposite side of the icon



8

migraine p2 — C2 (CERVICAL CONTRIBUTION panel)

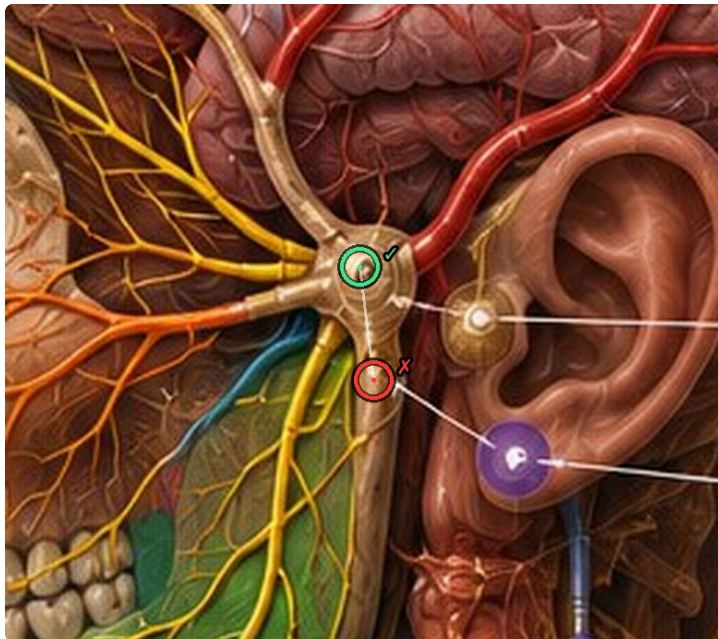
x ERASE (150,1264)

red posterior neck muscle, 16 px above the nerve

✓ DRAW AT (150,1280)

the yellow C2 ramus

The C3 arrow just below is correct and shows the intended form.



9

migraine p2 — F — Trigeminal ganglion (Gasserian)

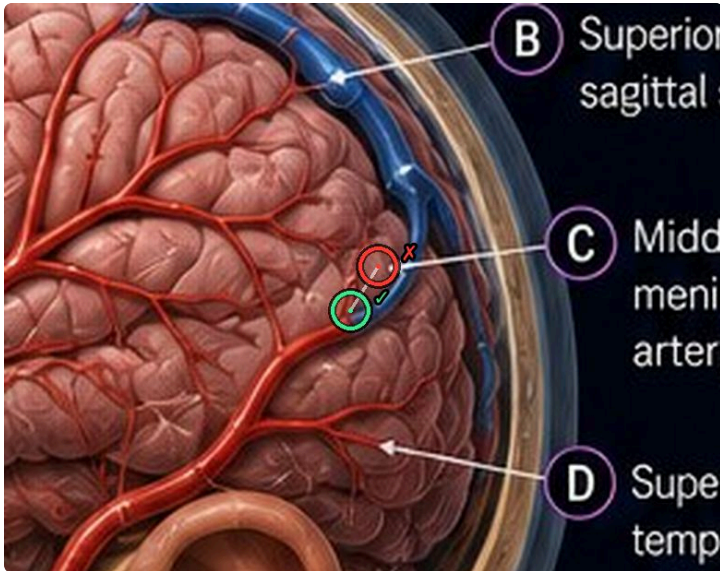
x ERASE (453,589)

the beige nerve trunk below the ganglion

✓ DRAW AT (448,548)

the ringed ganglion hub where V1/V2/V3 radiate

The leader currently passes over the purple marker sphere at (487,614) on its way.



10

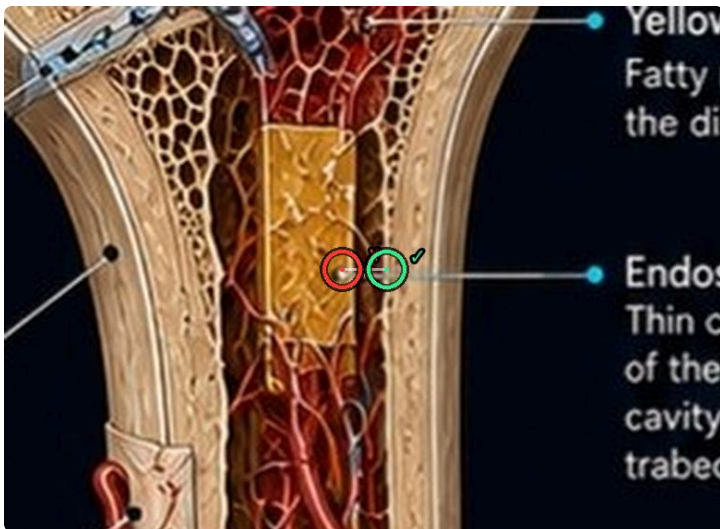
migraine p2 — C — Middle meningeal artery

☒ ERASE (598,412)

bare cortical gyri — no vessel of any kind under it

☒ DRAW AT (588,428)

the red arterial tree



11

osteomyelitis p2 — Endosteum

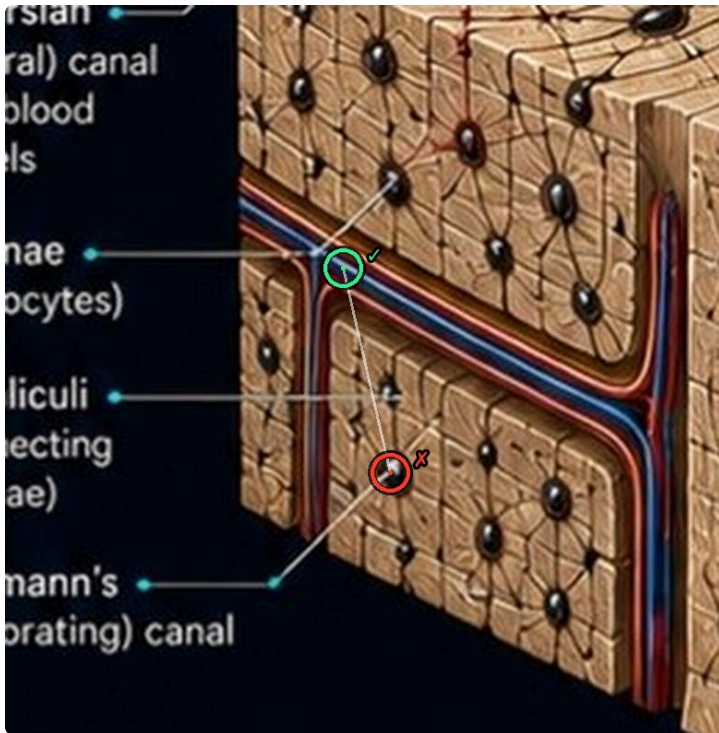
☒ ERASE (343,533)

inside the gold yellow-marrow block

☒ DRAW AT (359,533)

the inner surface of the cortical shell

At y=533 the layers are: gold marrow to x 346, dark transition 348-356, cortical shell 358-386.



12

osteomyelitis p2 — Volkmann's (perforating) canal

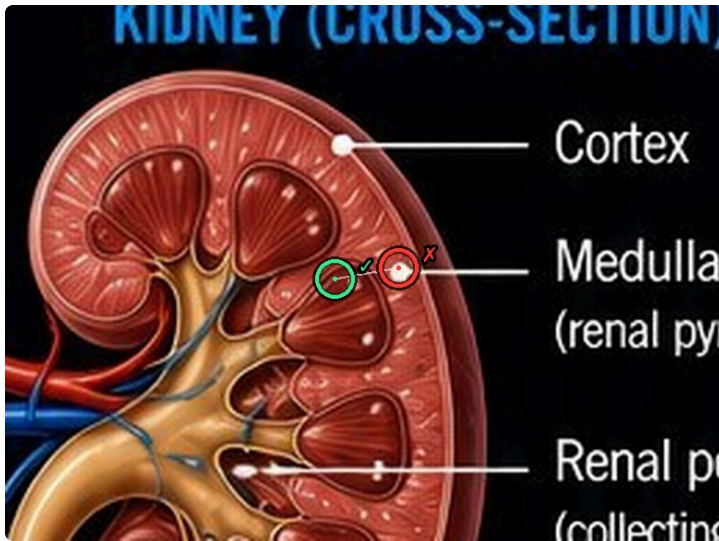
✗ ERASE (783,622)

a black almond-shaped osteocyte lacuna

✓ DRAW AT (766,548)

the transverse red/blue vessel pair crossing the cut face

“Lacunae” 110 px above uses the same dogleg on the same kind of target — that one is correct.



13

nephrolithiasis p2 — Medulla (renal pyramids)

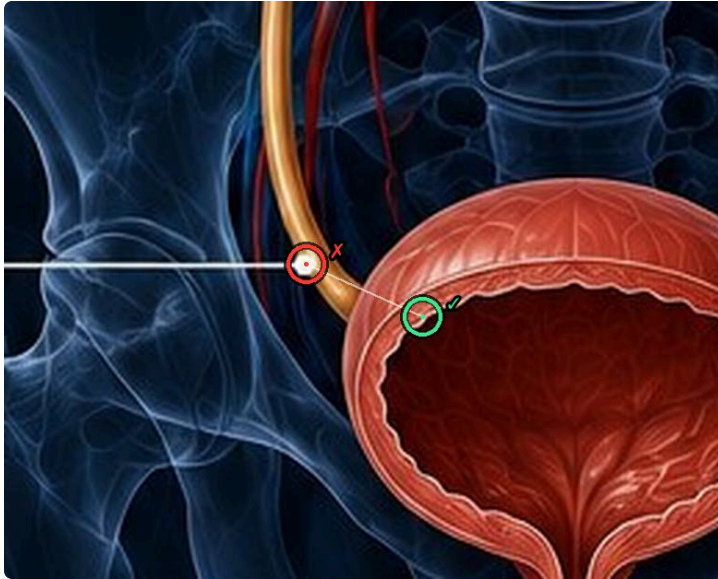
✗ ERASE (818,856)

the outer cortical band — the same band “Cortex” marks 51 px above

✓ DRAW AT (795,860)

inside a pyramid, in the striated tissue

The cortical band here is only 16 px wide, so the dot was placed carefully in the wrong layer.



14

nephrolithiasis p2 — Urinary bladder

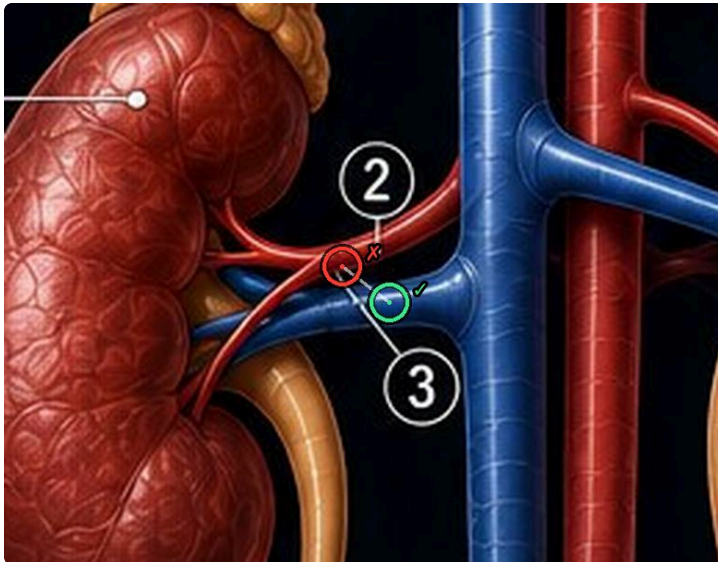
✗ ERASE (278,871)

the tan distal ureter

✓ DRAW AT (320,890)

inside the red bladder

| A 20 px miss between the two structures the page is about.



15

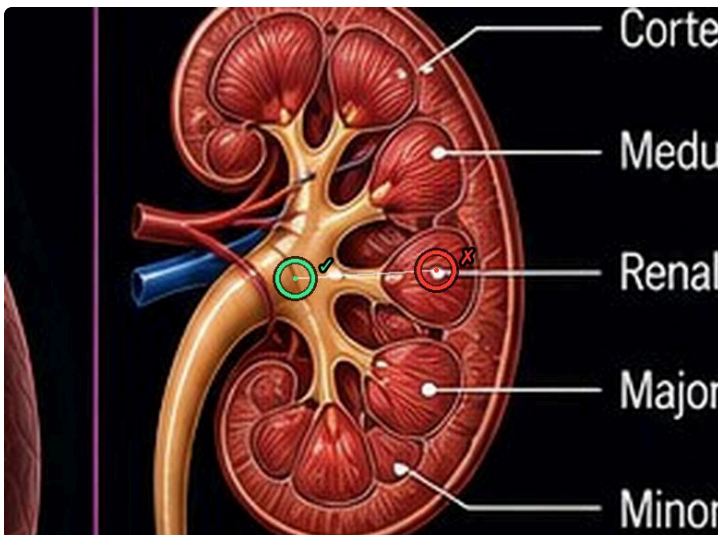
uti p2 — ③ Renal vein

✗ ERASE (413,413)

the red artery's lower edge — the vessel ② already names

✓ DRAW AT (430,426)

the blue renal vein, 12 px below



16

uti p2 — Renal pelvis (collecting system)

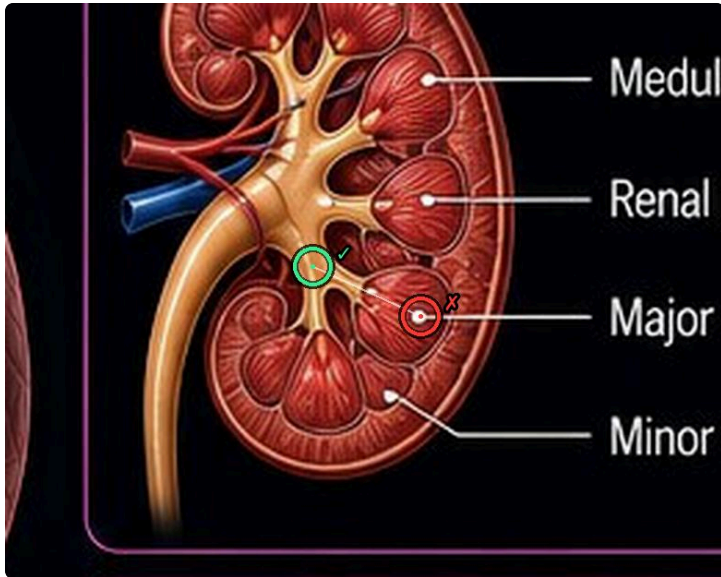
✗ ERASE (844,347)

a red medullary pyramid

✓ DRAW AT (793,350)

the tan funnel of the collecting system

| Three of five labels in this panel point at pyramids. Fix all three together.



17

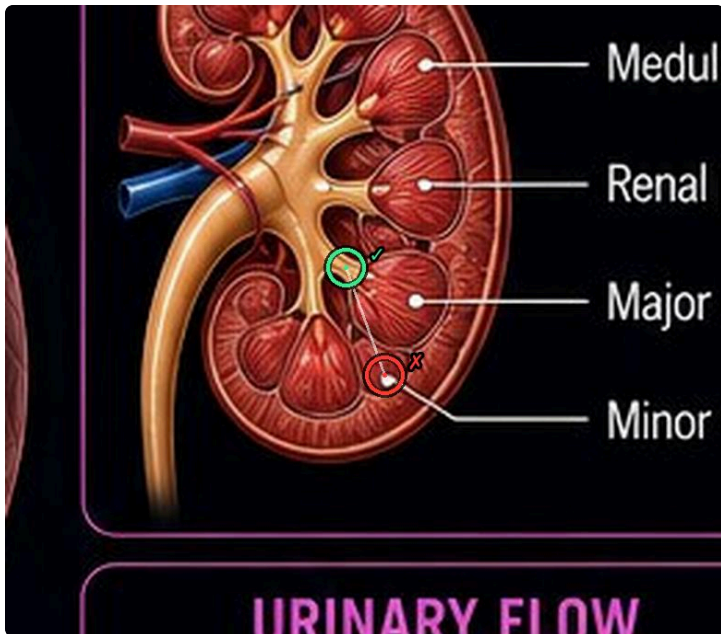
uti p2 — Major calyx

☒ ERASE (842,390)

a red medullary pyramid

☒ DRAW AT (803,372)

a tan branch of the pelvis



18

uti p2 — Minor calyx

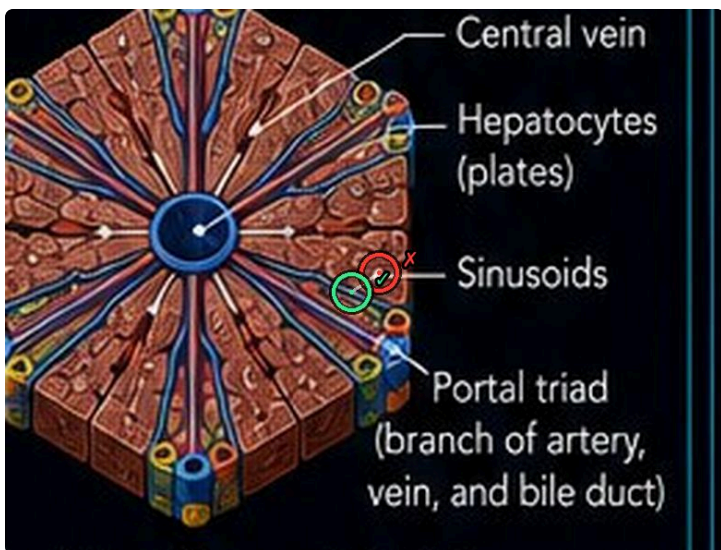
☒ ERASE (830,417)

the base of a third red pyramid

☒ DRAW AT (816,378)

a small tan cup around a papilla

The gold branches stop at about y 400 — nothing below that is collecting system.



19

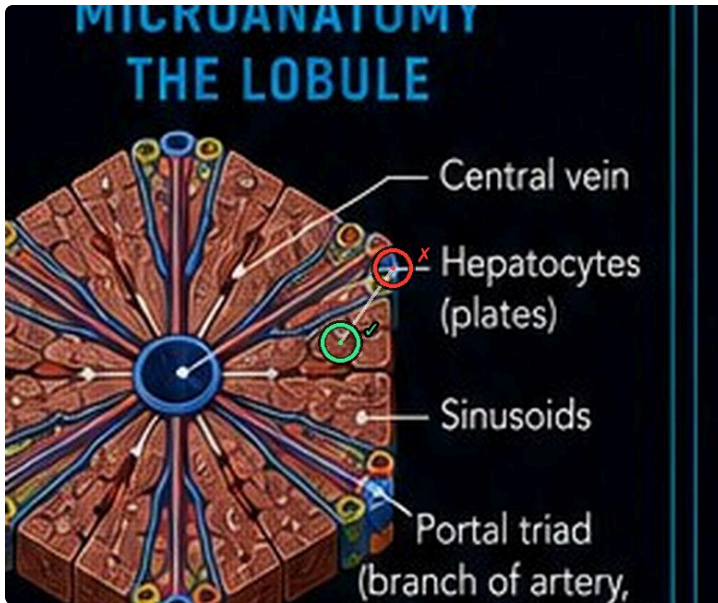
hepatitis p2 — Sinusoids (MICROANATOMY)

☒ ERASE (413,941)

inside a salmon hepatocyte plate block

☒ DRAW AT (403,948)

the blue radial channel between plates



20

hepatitis p2 — Hepatocytes (plates)

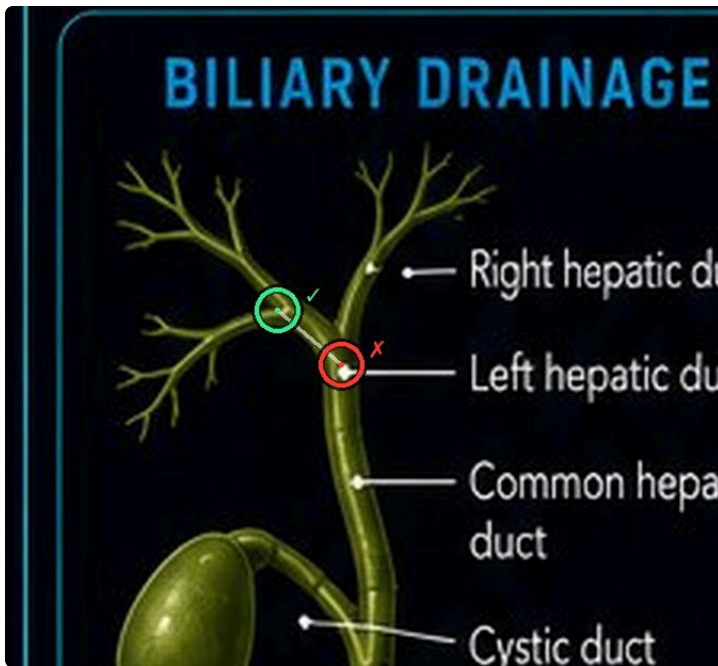
✗ ERASE (424,888)

the blue portal venule of the triad at the hexagon corner

✓ DRAW AT (405,915)

a salmon plate block

With a19, these two are effectively a swap: hepatocytes points at a vessel, sinusoids at hepatocytes.



21

hepatitis p2 — Left hepatic duct (BILIARY DRAINAGE)

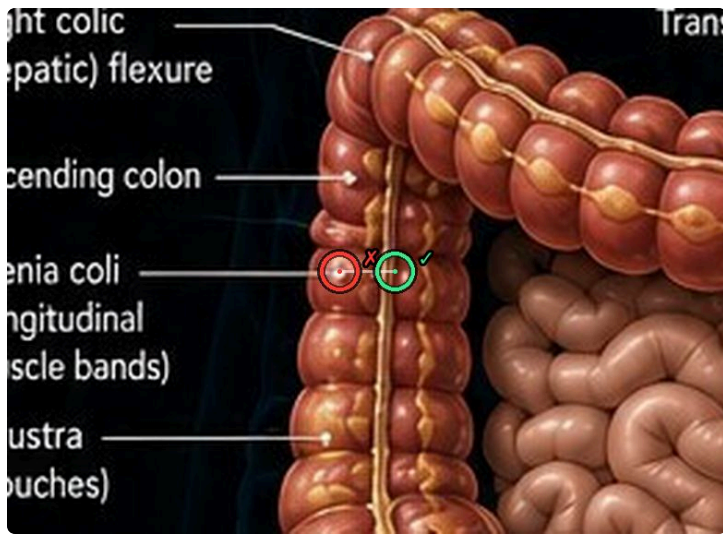
✗ ERASE (105,883)

the common hepatic trunk, 11 px below the confluence

✓ DRAW AT (85,866)

the left-going limb of the bifurcation

“Common hepatic duct” points at the same trunk 36 px lower.



22

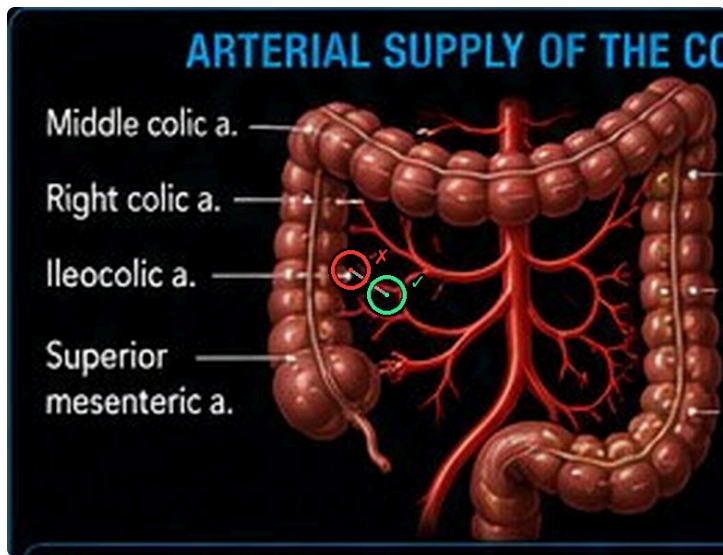
cdiff p2 — Taenia coli

✗ ERASE (169,366)

the free lateral haustral wall — what “Haustra” names 58 px below

✓ DRAW AT (189,366)

the tan longitudinal band, 19 px medial



23

cdiff p2 — Ileocolic a.

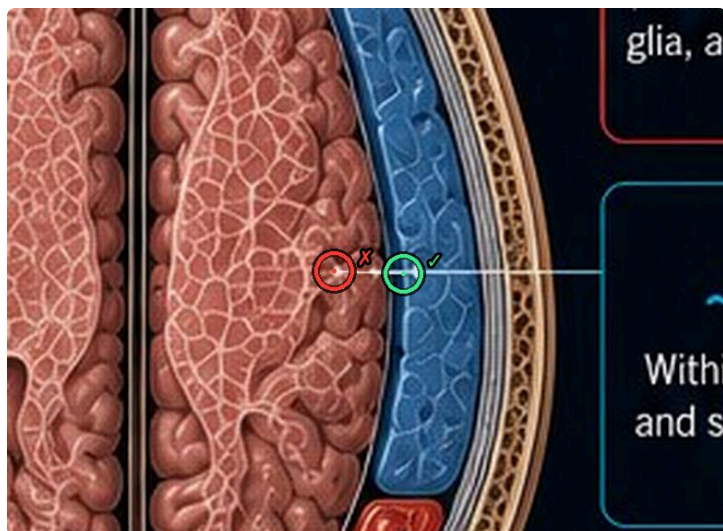
✗ ERASE (139,1143)

black background, 13 px short of the arcade

✓ DRAW AT (152,1152)

the arcade branch running to the cecum

|| Easiest erase on the page — pure black ground.



24

icp p2 — CSF ~10%

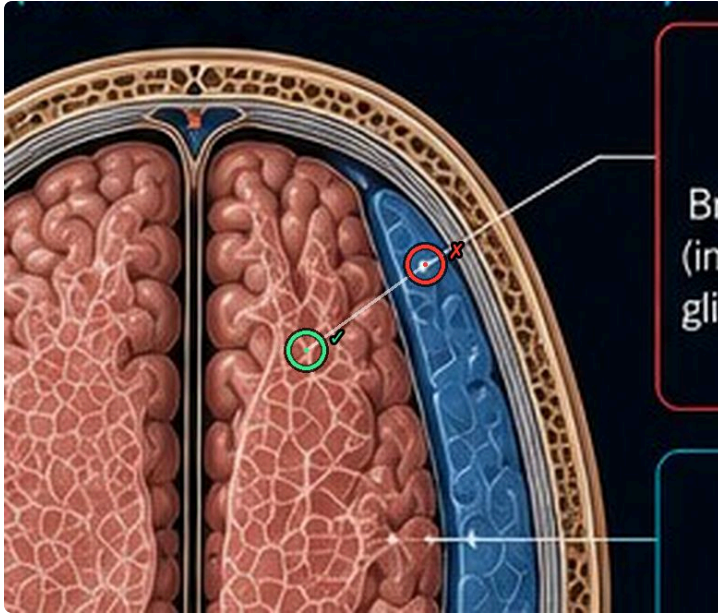
✗ ERASE (770,468)

pink brain parenchyma — the arrow crosses the whole blue band and stops 22 px past it

✓ DRAW AT (795,469)

the blue CSF band

|| At y=469 the boundary is x≈790: pink to 788, blue from 792.



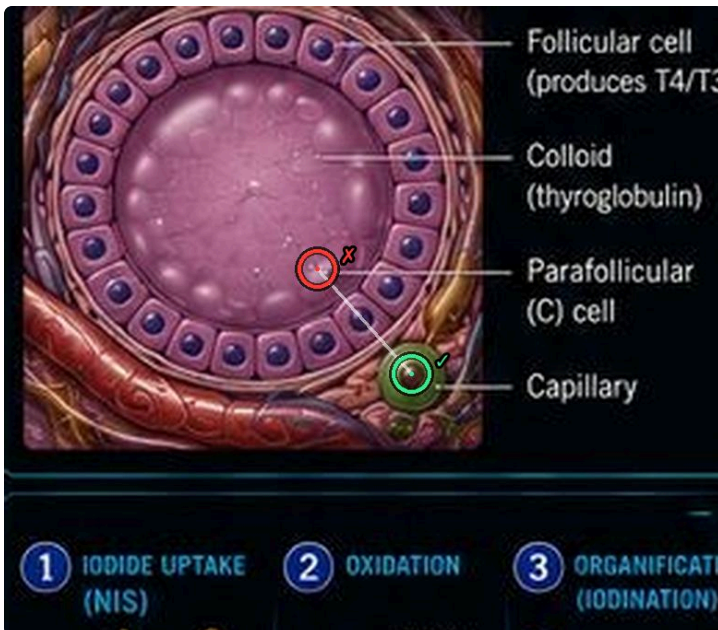
25

icp p2 — **BRAIN ~80%**

✗ ERASE (783,369)
inside the blue CSF band

✓ DRAW AT (740,400)
the pink brain parenchyma

| With a24 this is a straight mutual swap on the Monro-Kellie panel.



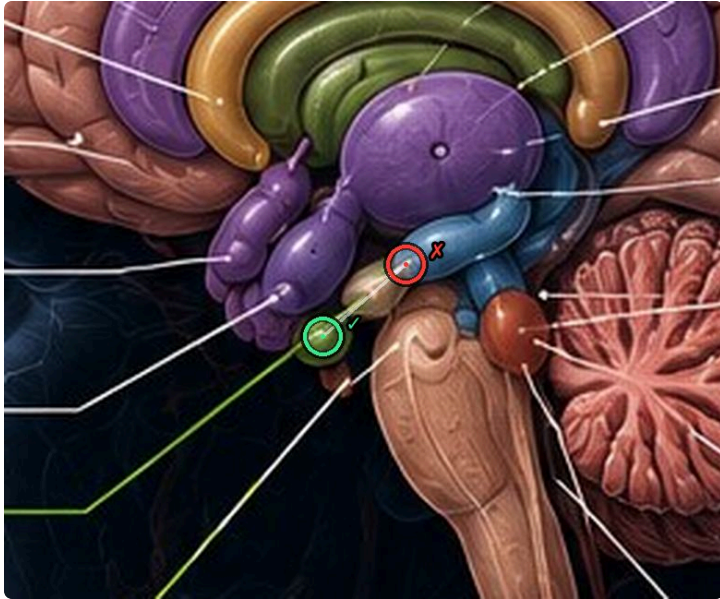
26

hypothyroid p2 — **Parafollicular (C) cell**

✗ ERASE (126,672)
the pale colloid at the follicle centre

✓ DRAW AT (160,710)
the olive-green C cell outside the epithelium

| 800×1200 page. The panel's point is that C cells sit OUTSIDE the follicle.



27

parkinsons p2 — Subthalamic nucleus (STN)

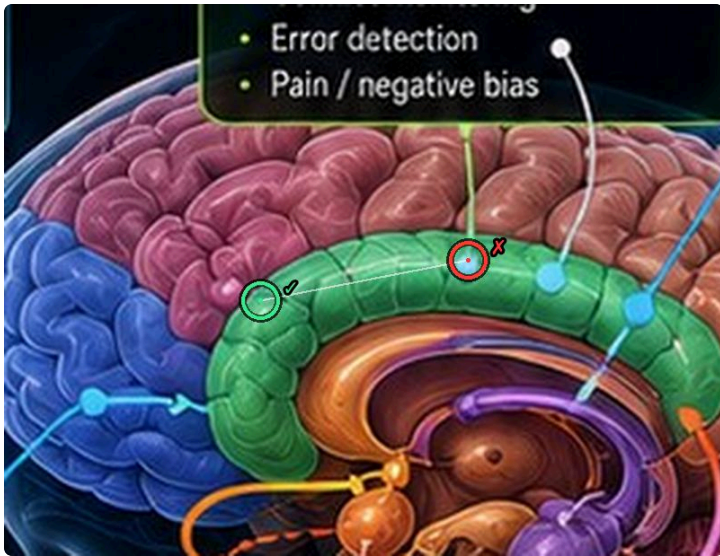
✗ ERASE (347,540)

the tan wedge, 33 px past the sphere

✓ DRAW AT (317,566)

the olive-green sphere — the colour of the label's own ink

| The leader passes straight over the green sphere on its way.



28

suicide p2 — Anterior Cingulate Cortex (ACC)

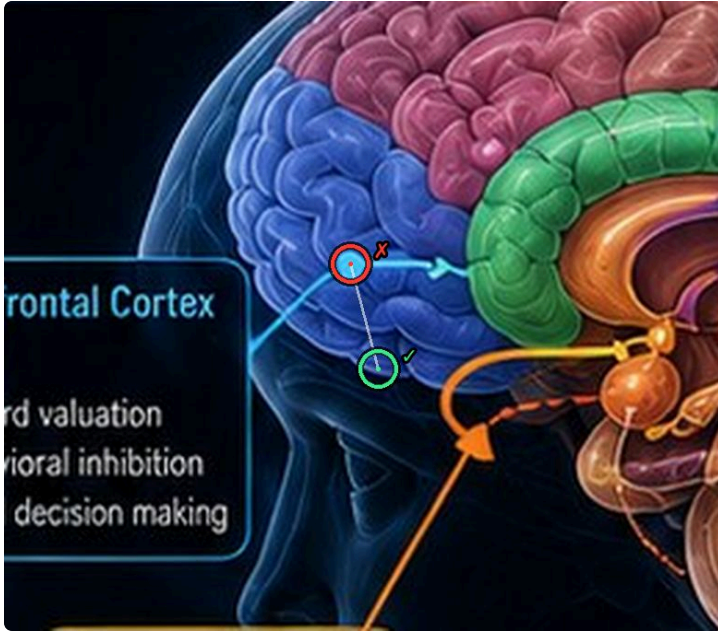
✗ ERASE (339,455)

the green cingulate band's midpoint — mid-cingulate, not anterior

✓ DRAW AT (262,470)

the rostral limb of the same band

| The band runs x 233–440, so the current dot is its exact midpoint.



29

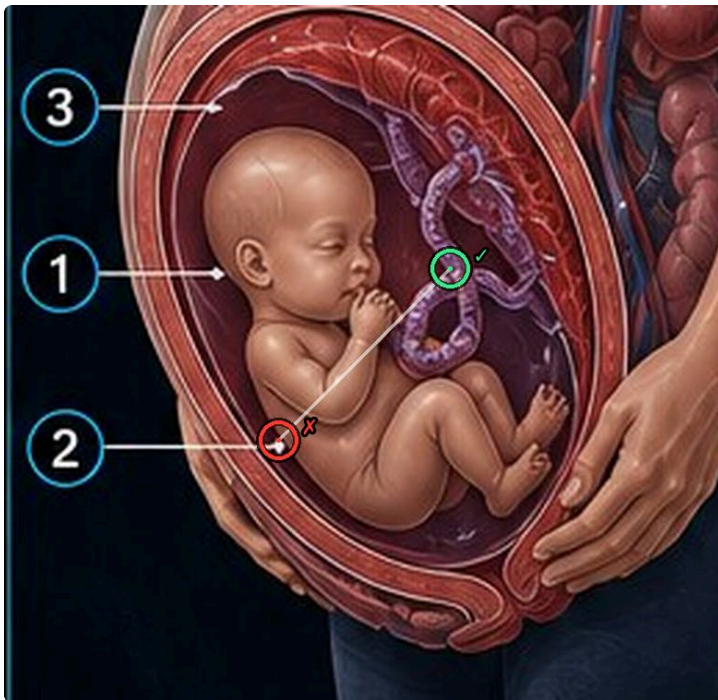
suicide p2 — Orbitofrontal Cortex (OFC)

✗ ERASE (200,507)

mid medial-frontal cortex, 40 px above the orbital surface

✓ DRAW AT (210,545)

the orbital (inferior) surface of the blue frontal lobe



30

gdm p2 — ② UMBILICAL CORD

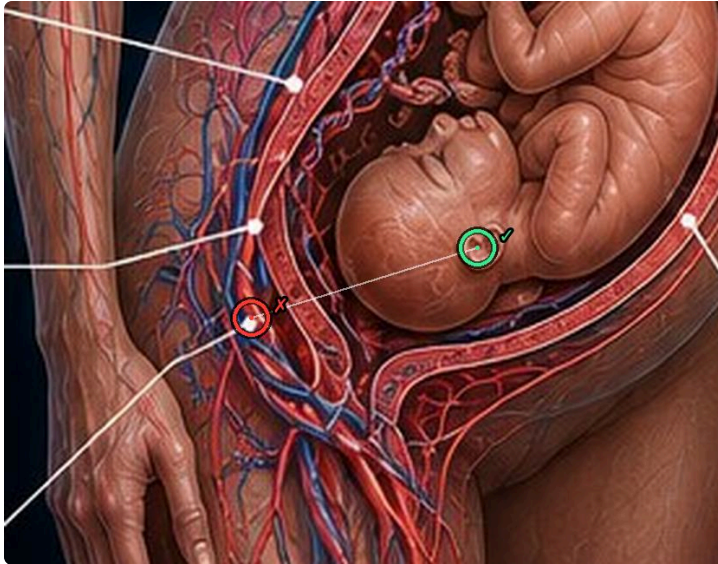
✗ ERASE (373,762)

the fetal back / buttock

✓ DRAW AT (435,700)

the purple coiled cord

The cord is the only purple structure in the uterus.



31

preeclampsia p2 — UMBILICAL CORD

✗ ERASE (285,879)

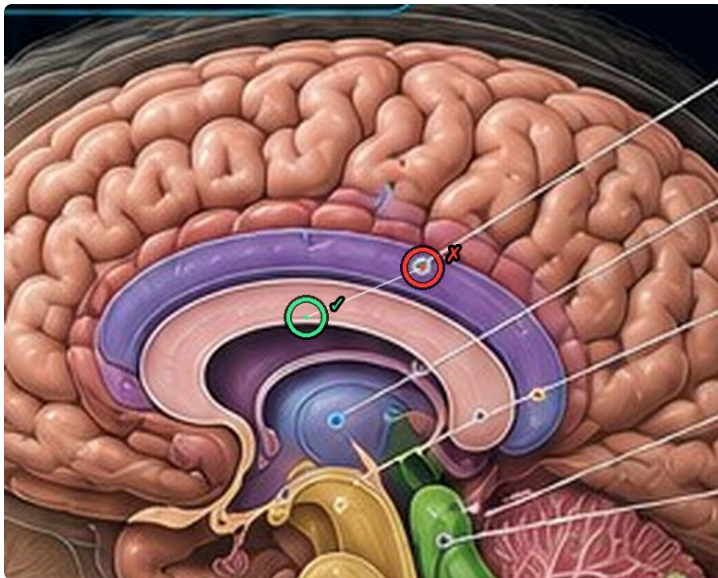
maternal pelvic vessels, outside the uterus

✓ DRAW AT (372,852)

the cord's insertion at the fetal abdomen

B Group B — move one dot, textured background

Same job, but the old terminator sits on gyri, folia, trabeculae or bowel wall. Clone stamp with the grain — see the method notes at the front.



32

dementia p2 — Corpus callosum

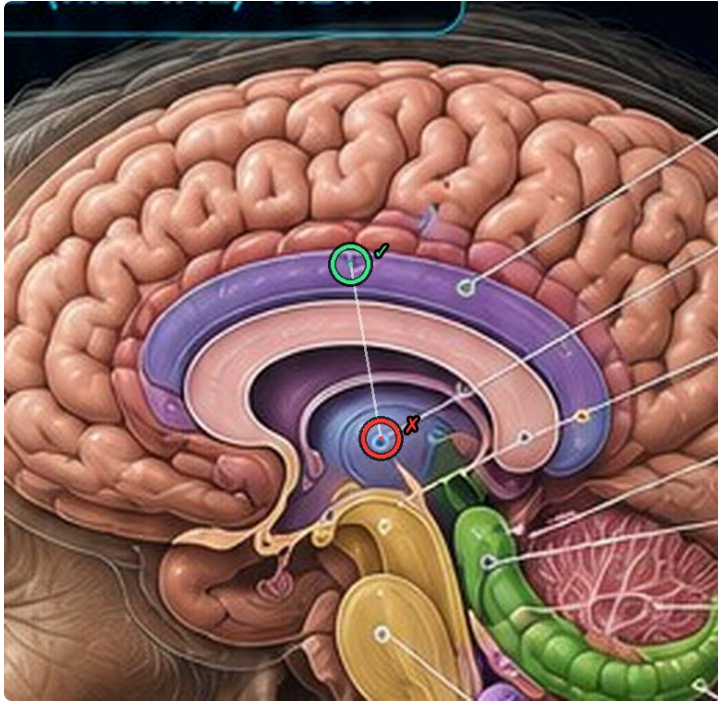
✗ ERASE (292,424)

the purple cingulate arc

✓ DRAW AT (250,442)

the salmon callosal band inside it

Rows b01–b03 are a three-way rotation: each label stands on the structure the label above it names.



33

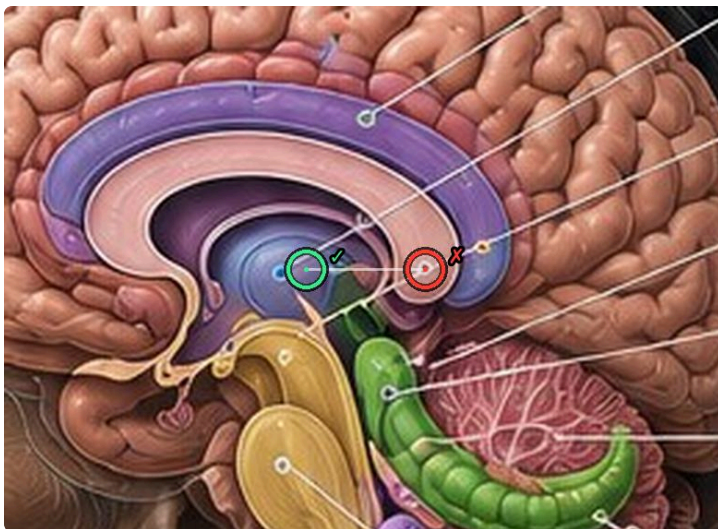
dementia p2 — Cingulate gyrus

☒ ERASE (261,478)

the blue thalamus

☒ DRAW AT (250,415)

the purple arc above the corpus callosum



34

dementia p2 — Thalamus

☒ ERASE (313,478)

the salmon corpus callosum, posterior third

☒ DRAW AT (270,478)

the blue oval



35

dementia p2 — Amygdala

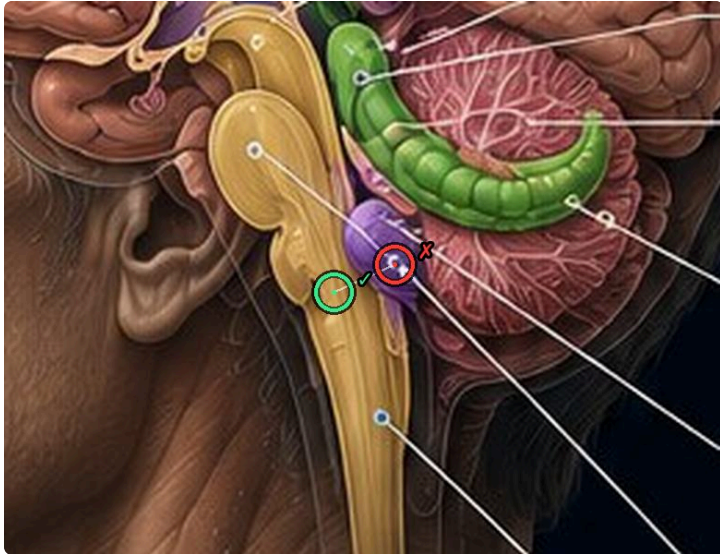
☒ ERASE (263,549)

the middle of the gold pons

☒ DRAW AT no coordinate — place from the source

the anterior medial temporal region

No target coordinate — place it from the source. Sampled (202,171,132) at the dot = pons.



36

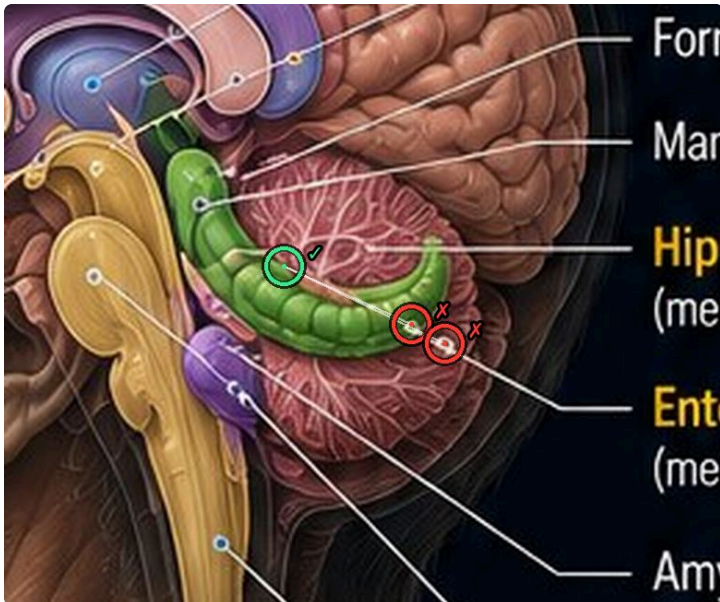
dementia p2 — Brainstem

✗ ERASE (312,590)

a purple medial-temporal blob behind the brainstem

✓ DRAW AT (290,600)

the gold column



37

dementia p2 — Hippocampus (sagittal)

✗ ERASE (376,566) and (388,573)

TWO ring terminators straddling the green hippocampus rim and the pink cerebellum

✓ DRAW AT (330,545)

one ring, well inside the green body

Erase both rings. A leader with two endpoints gives the reader two answers.

38

dementia p2 — Lateral ventricles (CORONAL)

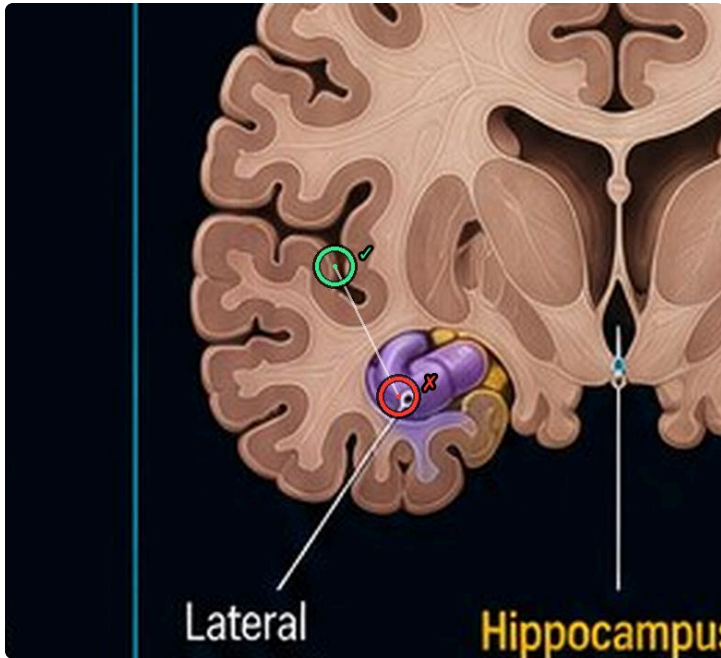
x ERASE (730,472)

a purple hippocampus

✓ DRAW AT (707,425)

a dark ventricular cavity

On this panel all three labels are wrong — see b07-b09.



39

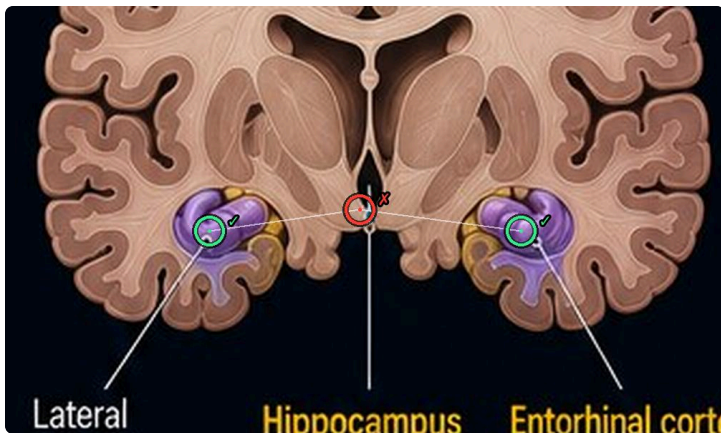
dementia p2 — Hippocampus (bilateral)

x ERASE (805,458)

the midline, at third-ventricle level

✓ DRAW AT (735,468) and (880,468)

the two purple bodies, one leader each



40

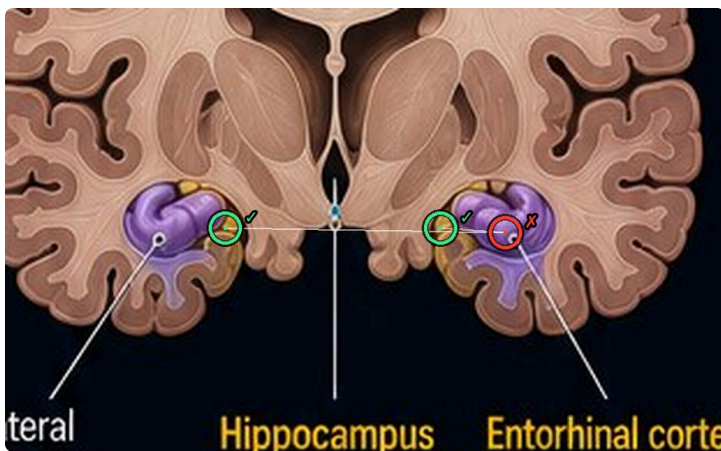
dementia p2 — Entorhinal cortex (bilateral)

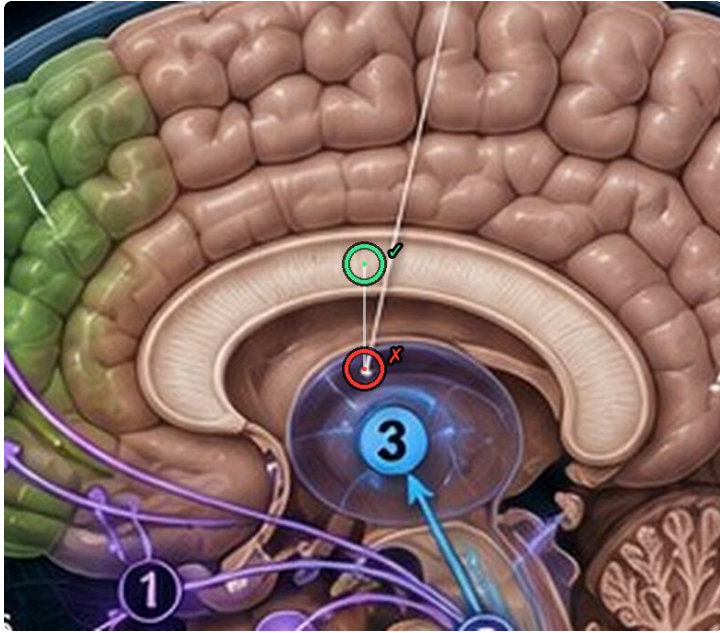
x ERASE (883,470)

a purple hippocampus

✓ DRAW AT (762,468) and (855,468)

the gold/ochre bands medial and inferior to each hippocampus





41

schizophrenia p2 — Corpus Callosum

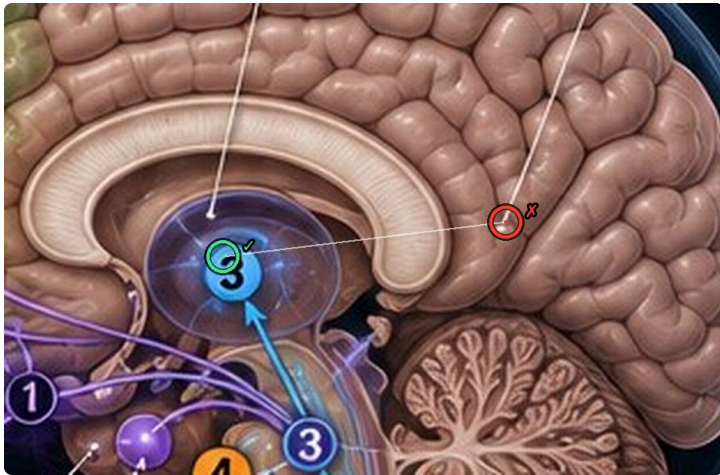
✗ ERASE (489,424)

inside the blue thalamic mass

✓ DRAW AT (489,386)

the pale ribbed callosal band, straight up the same leader

|| The leader already passes through the band on its way down.



42

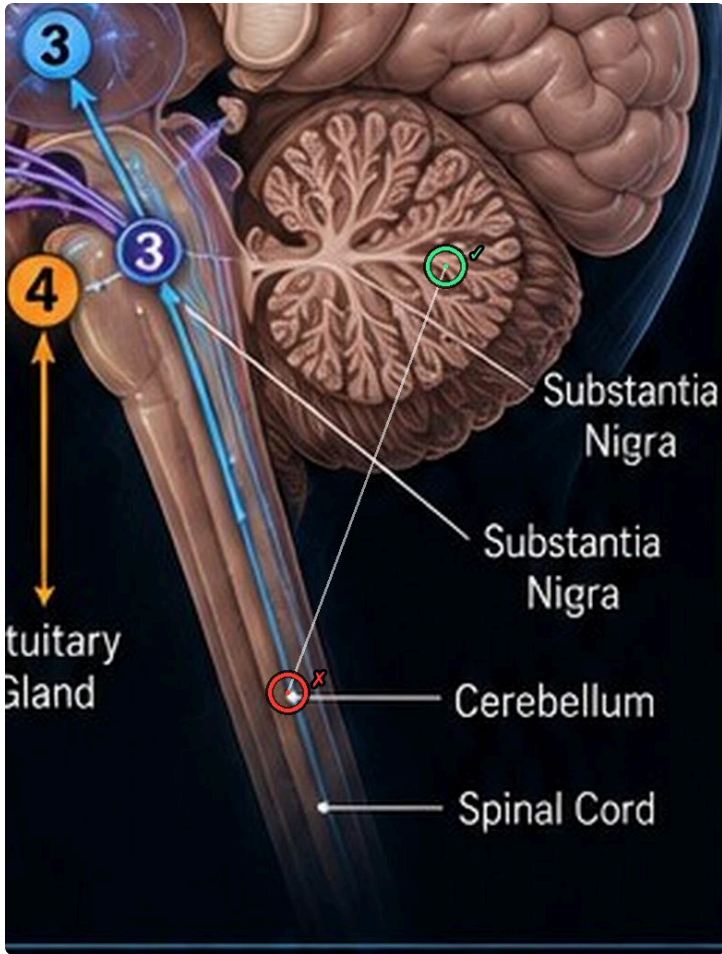
schizophrenia p2 — Thalamus

✗ ERASE (618,428)

parietal cortex, 124 px from the thalamus

✓ DRAW AT (495,443)

the blue thalamic mass



43

schizophrenia p2 — Cerebellum

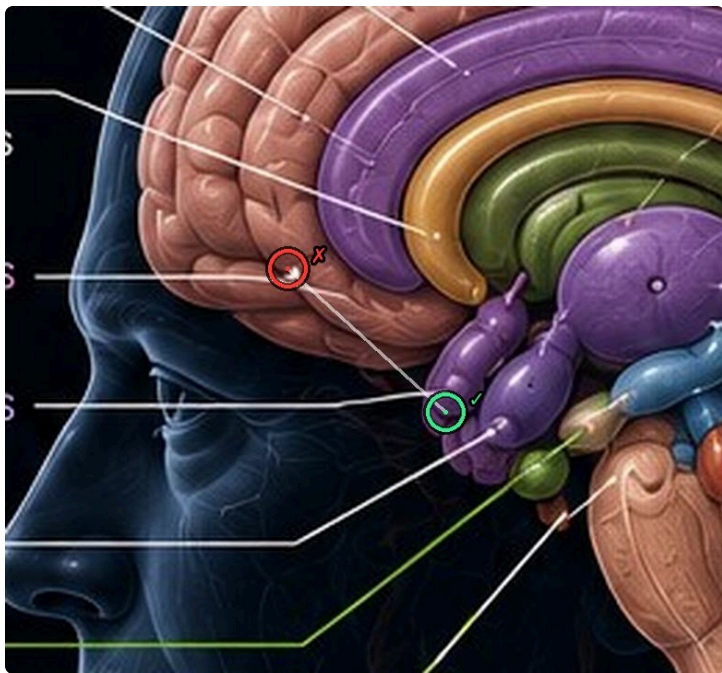
✗ ERASE (583,684)

the brainstem / upper cord tube

✓ DRAW AT (640,530)

the foliated cerebellum

“Spinal Cord” correctly points at the same tube 39 px lower.



44

parkinsons p2 — Globus pallidus (external)

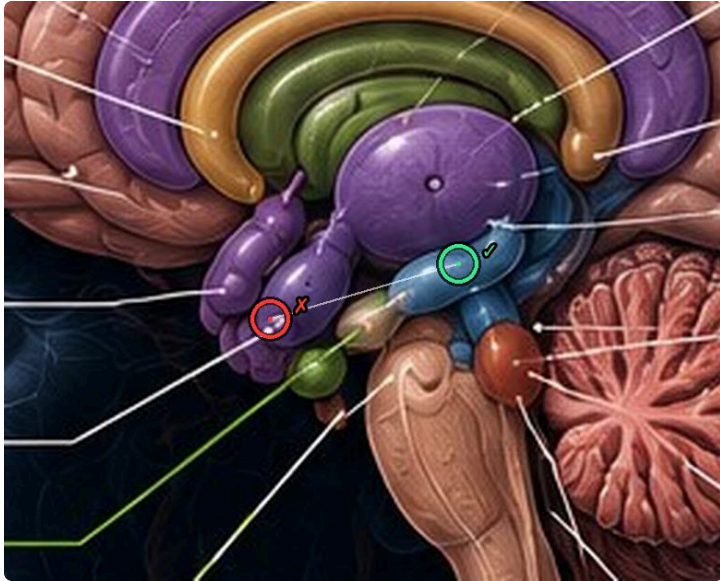
✗ ERASE (226,493)

pink frontal cortex — outside every coloured deep-grey structure

✓ DRAW AT (283,545)

the purple lump chain

“Globus pallidus (internal)” at (283,537) is correct and shows the target.



45

parkinsons p2 — Thalamus (left column)

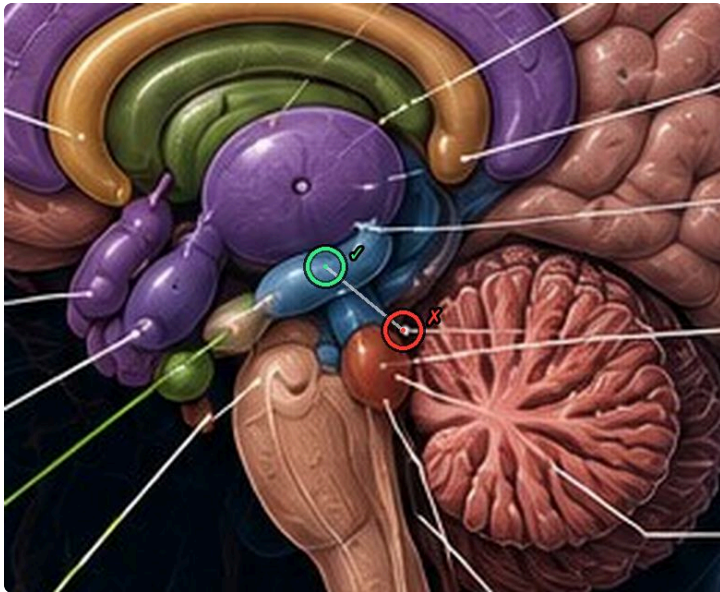
✗ ERASE (300,548)

the purple pallidal lumps, 21 px from the GPi dot

✓ DRAW AT (368,528)

the blue body — the key's thalamus colour

The coronal panel's key assigns blue to thalamus and purple to pallidum.



46

parkinsons p2 — Thalamus (right column)

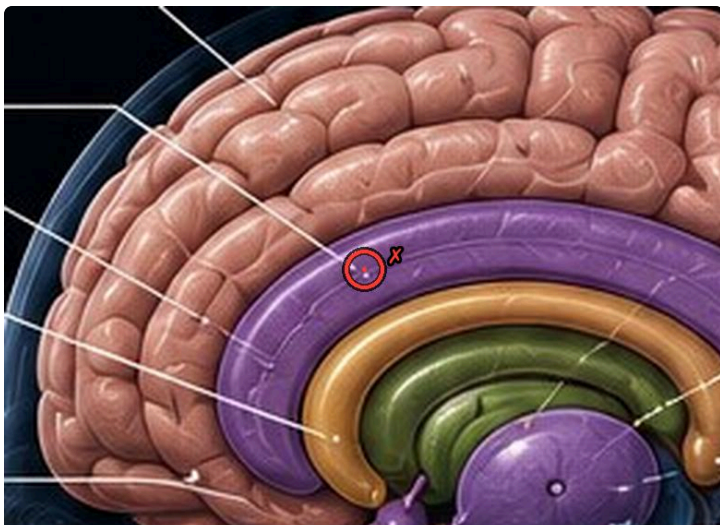
✗ ERASE (396,551)

the red nucleus's upper-right edge

✓ DRAW AT (368,528)

the blue body

Three leaders converge on the red nucleus. Only "Red nucleus" belongs there.



47

parkinsons p2 — Cingulate gyrus

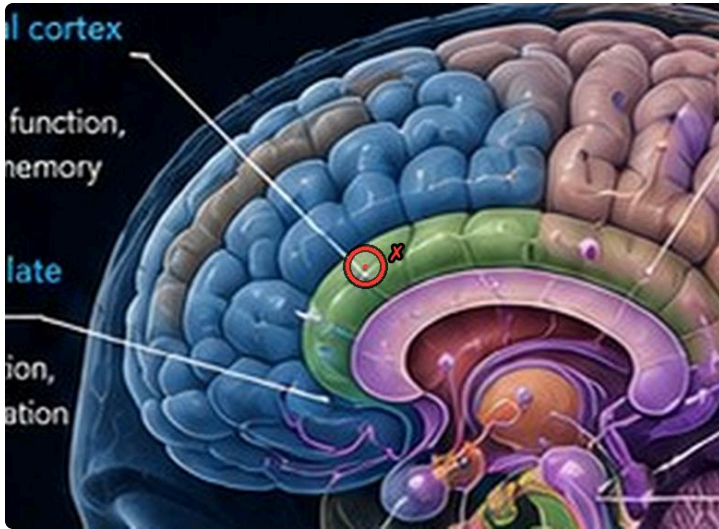
✗ ERASE (290,420)

the purple callosal band — where "Corpus callosum" ends 43 px away

✓ DRAW AT *no coordinate — place from the source*

the pink cortical ribbon immediately outside the arc

About 15 px up and out.



48

depression p2 — DLPFC

✗ ERASE (473,323)

the GREEN cingulate band — the colour the legend assigns to “Cingulate cortex”

✓ DRAW AT no coordinate — place from the source the blue frontal convexity above and in front of the band

| Sampled (152,176,132) at the dot = the green band.



49

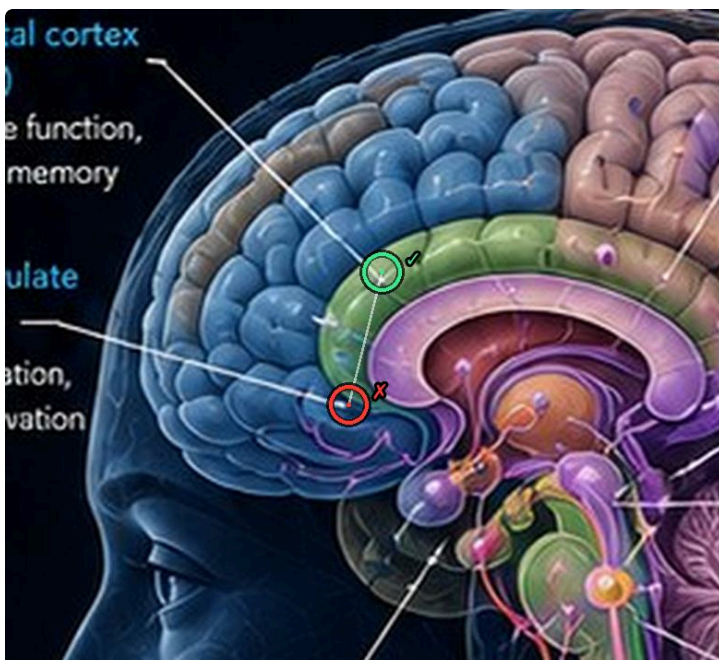
depression p2 — vmPFC

✗ ERASE (577,323)

the same band's posterior/olive end — high and far back

✓ DRAW AT (465,375)

ventral-anterior frontal cortex, low and forward



50

depression p2 — Anterior cingulate cortex (ACC)

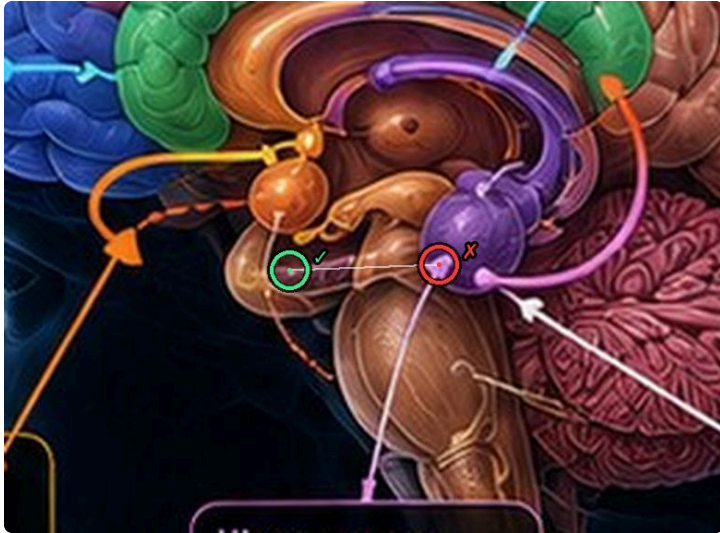
✗ ERASE (461,371)

blue frontal cortex

✓ DRAW AT (473,323)

the green cingulate band — where the DLPFC leader currently ends

| b17-b19 are an exchange: two prefrontal labels on the cingulate, the cingulate label on prefrontal cortex.



51

suicide p2 — Hippocampus

✗ ERASE (359,578)

the violet raphe sphere — 33 px from the Raphe Nuclei arrowhead on the same ball

✓ DRAW AT (305,580)

the plum medial-temporal band

Sampled (123,56,73) at the target — distinct from the orange amygdala and the surrounding brown.



52

suicide p2 — Amygdala

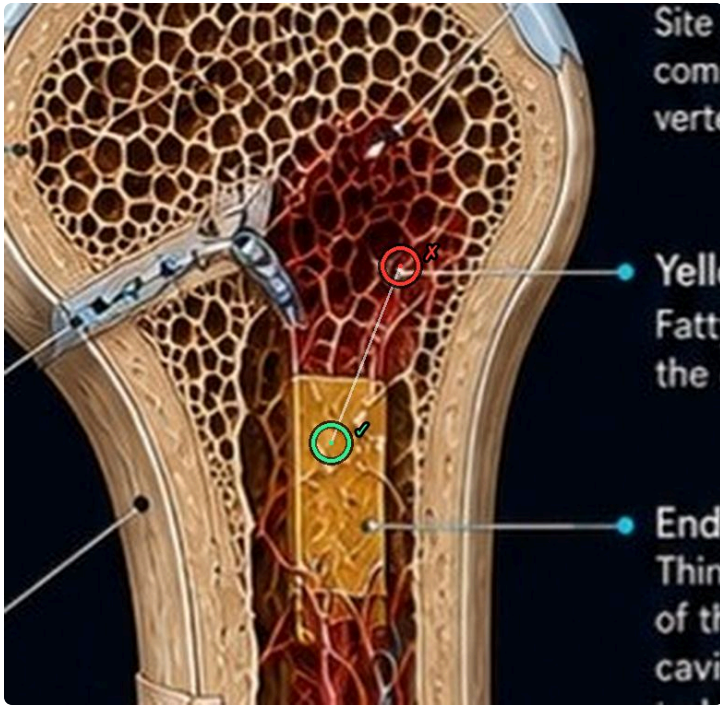
✗ ERASE (245,571)

empty dark background below the frontal lobe

✓ DRAW AT (305,556)

the orange amygdala sphere

Its leader is the same orange as the circuit arrows, which is why it reads as deliberate.



53

osteomyelitis p2 — Yellow marrow

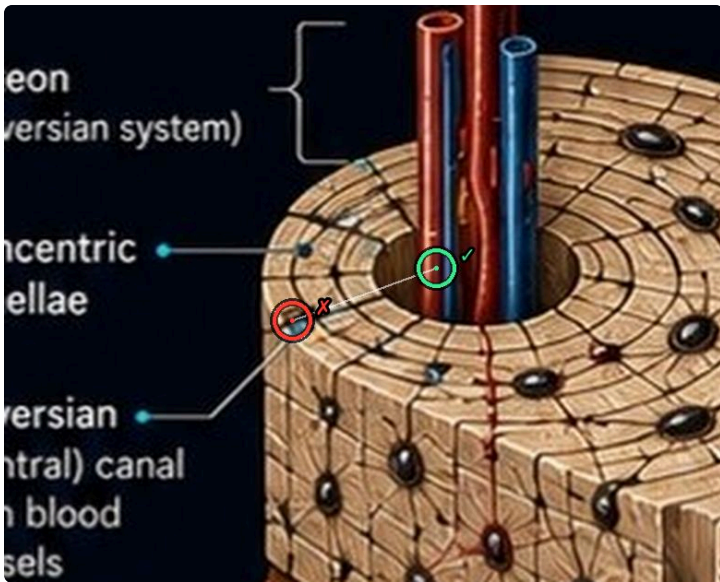
ERASE (353,441)

deep red marrow of the metaphysis

DRAW AT (328,505)

the gold fatty-marrow block (x 313-347, y 478-560)

“Red marrow” above it is correct, so these are not swapped — yellow marrow simply stops one structure short.



54

osteomyelitis p2 — Haversian (central) canal

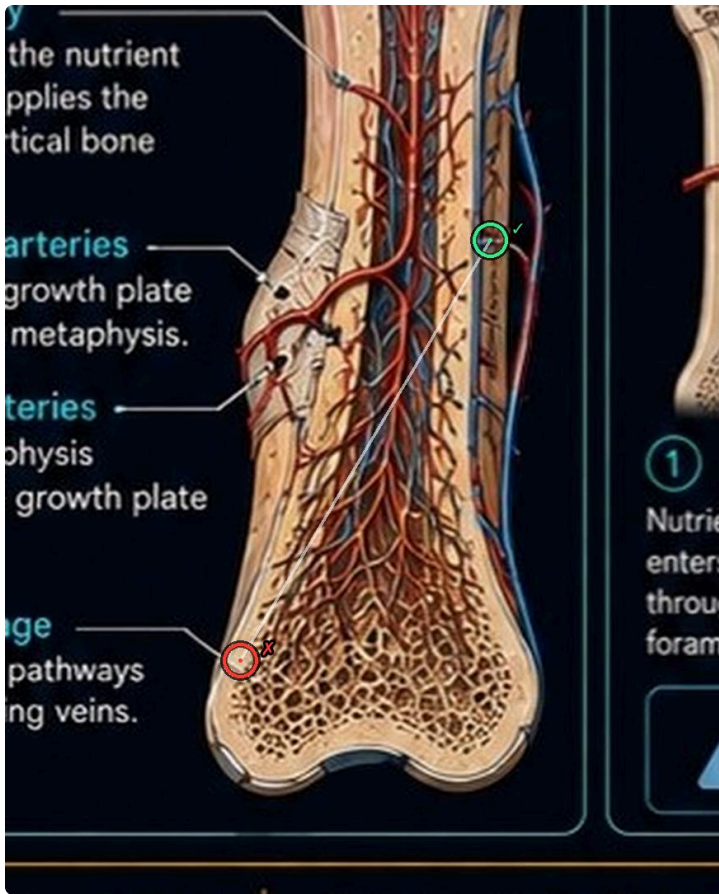
ERASE (738,421)

the concentric lamellar rings, 29 px from where “Concentric lamellae” points

DRAW AT (790,402)

inside the dark central opening with the vessels

Aimed at the canal, stops 27 px short.



55

osteomyelitis p2 — Venous drainage

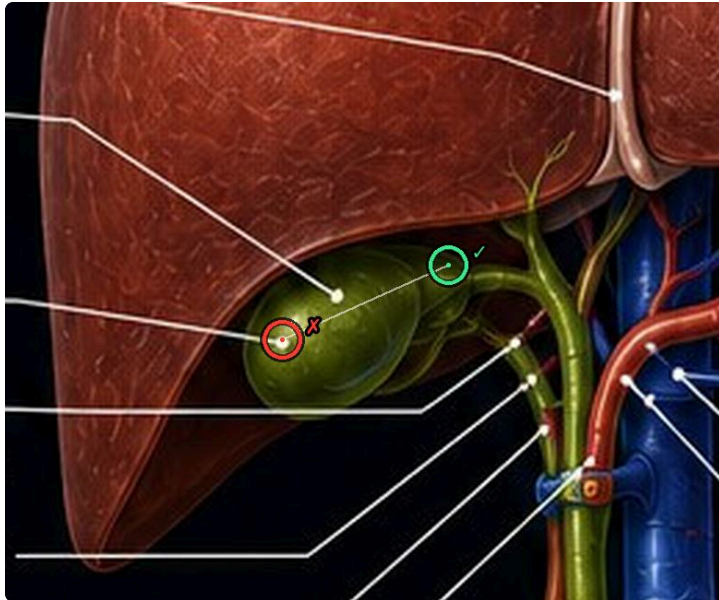
✗ ERASE (285,1250)

the distal cortical shell — no vein at the endpoint

✓ DRAW AT (386,1080)

the medullary venous channel

Careful: the bone's own outline carries a pale blue-grey highlight at x 391-405 that looks like a vein and is not.



56

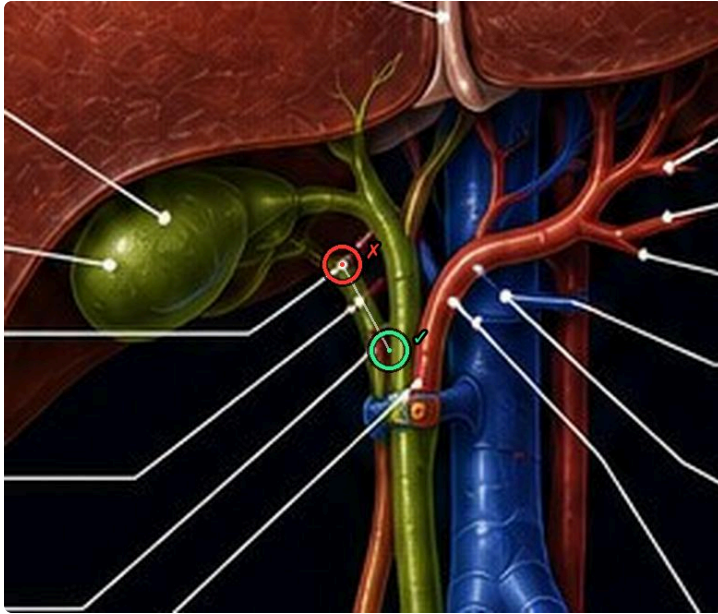
hepatitis p2 — Cystic duct

✗ ERASE (265,523)

inside the gallbladder sac — where "Gallbladder" points 22 px away

✓ DRAW AT (325,496)

the green tube leaving the sac's neck



57

hepatitis p2 — Common bile duct

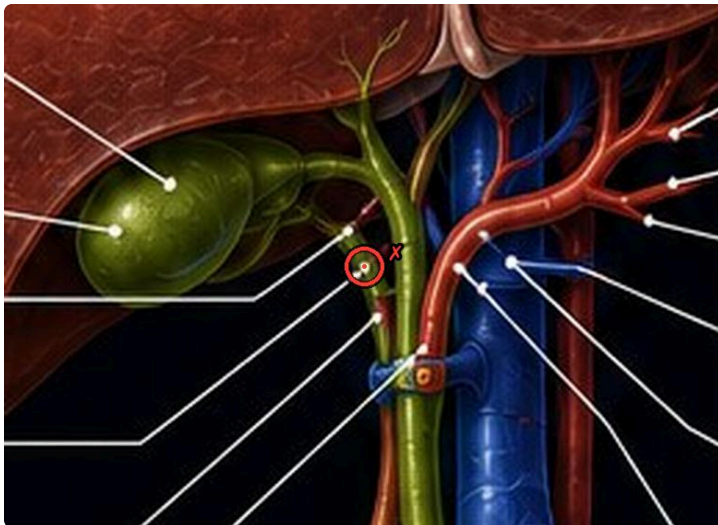
☒ **ERASE** (349,524)

the green tree, but ABOVE the cystic-duct confluence

☒ **DRAW AT** (366,555)

the green trunk below the junction

Above that junction the tube is the common hepatic duct.



58

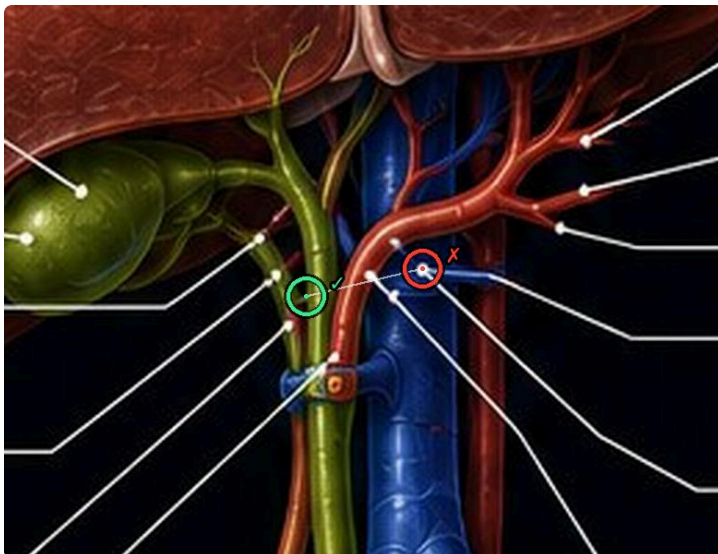
hepatitis p2 — Right hepatic artery

☒ **ERASE** (355,537)

the green bile duct — sampled (90,94,39)

☒ **DRAW AT** no coordinate — place from the source

the dark maroon arterial twig ~5 px up-right, clear of the duct



59

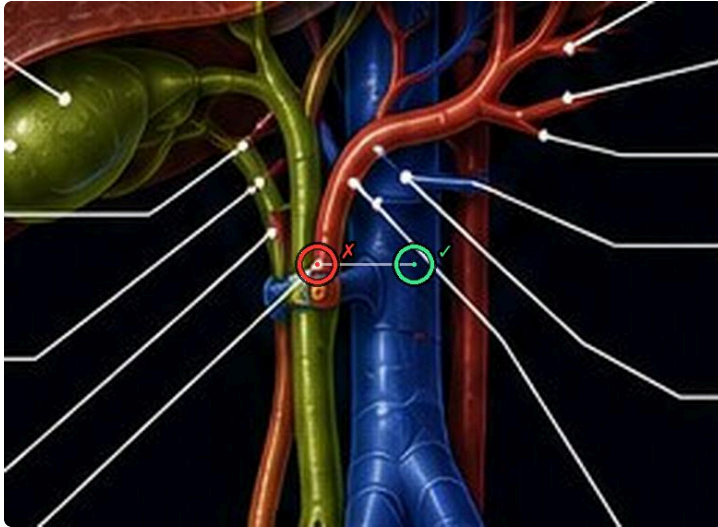
hepatitis p2 — Bile duct (from L and R hepatic ducts)

☒ **ERASE** (408,535)

the blue IVC — sampled (30,41,84)

☒ **DRAW AT** (366,545)

the green biliary trunk, 42 px left



60

hepatitis p2 — Inferior vena cava

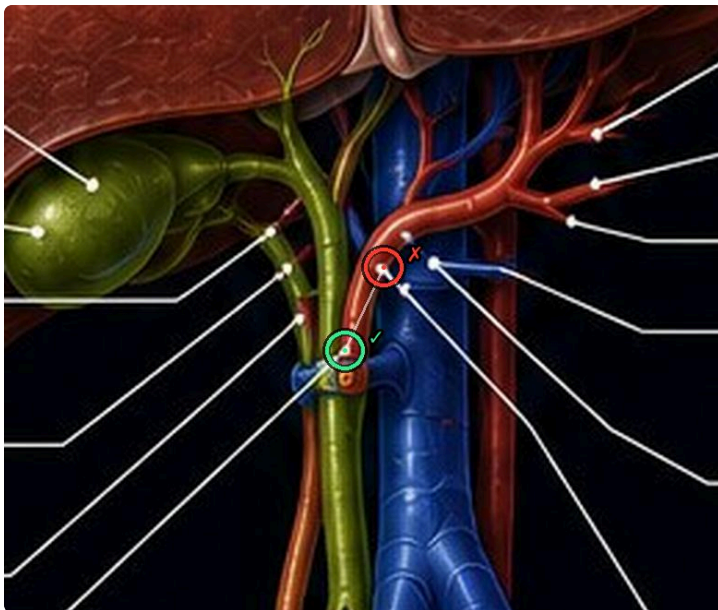
✗ ERASE (376,567)

the small orange portal-triad widget

✓ DRAW AT (411,567)

the blue IVC trunk, 35 px right

Do this one together with b30 — it is a clean exchange.



61

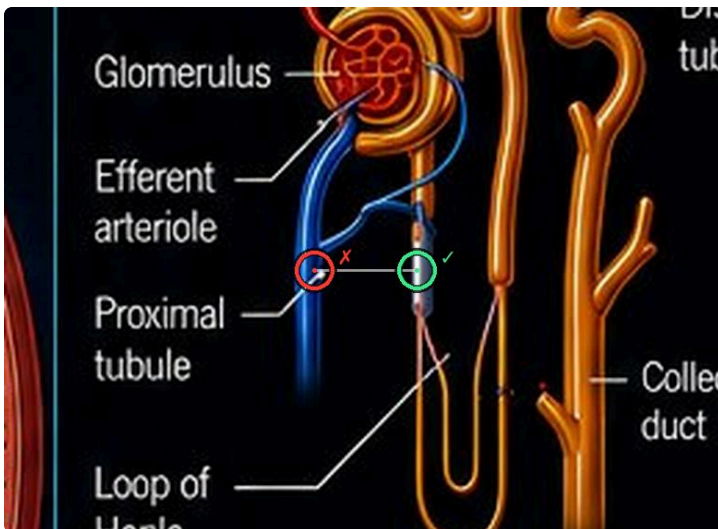
hepatitis p2 — Hepatic portal triad

✗ ERASE (390,537)

the bright red hepatic artery alone — one of the three, not the bundle

✓ DRAW AT (376,567)

the triad widget the IVC label vacates



62

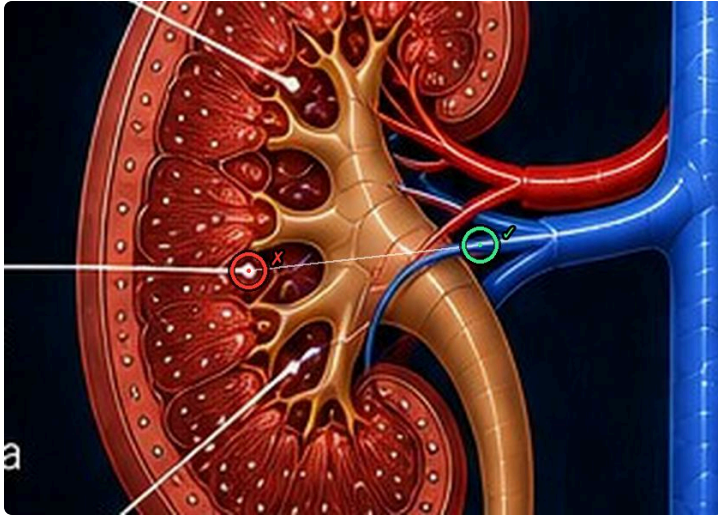
ckd p2 — Proximal tubule

✗ ERASE (821,431)

the blue efferent arteriole / peritubular vessel

✓ DRAW AT (858,431)

the orange-walled tubule, 37 px right



63

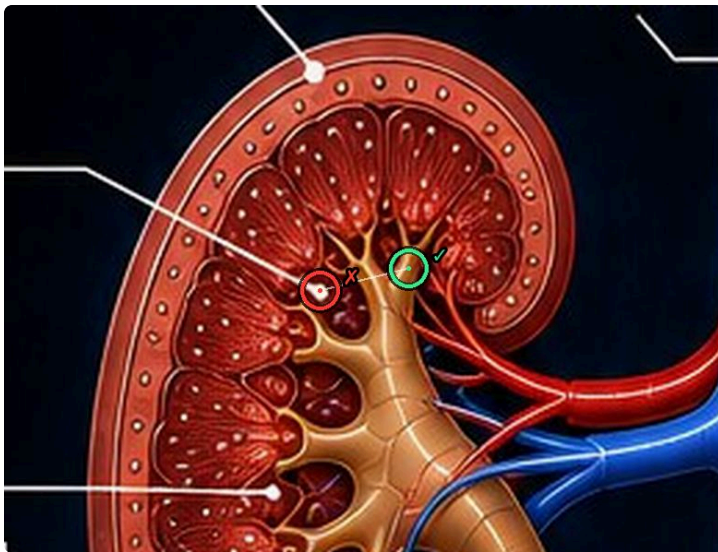
ckd p2 — RENAL VEIN

☒ ERASE (275,510)

kidney parenchyma — sampled (110,35,34)

☒ DRAW AT (365,500)

the blue renal vein at the hilum, sampled (31,56,117)



64

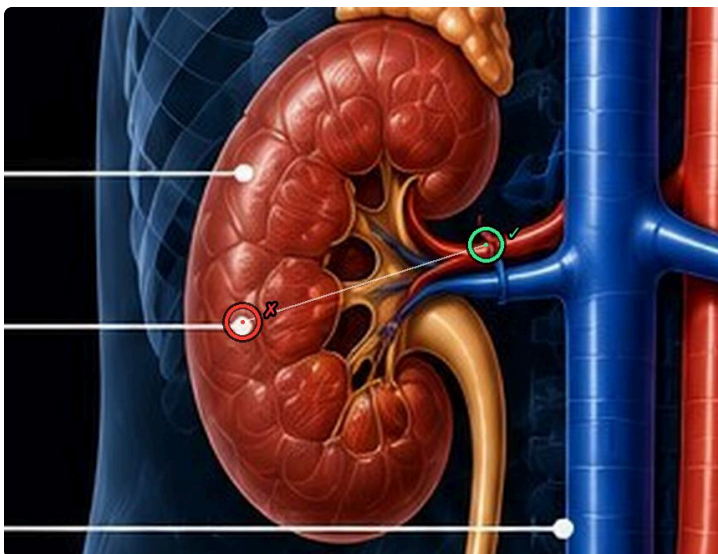
ckd p2 — RENAL ARTERY

☒ ERASE (292,437)

a medullary pyramid inside the renal sinus — no vessel at all

☒ DRAW AT (324,429)

the red arterial trunk at the hilum



65

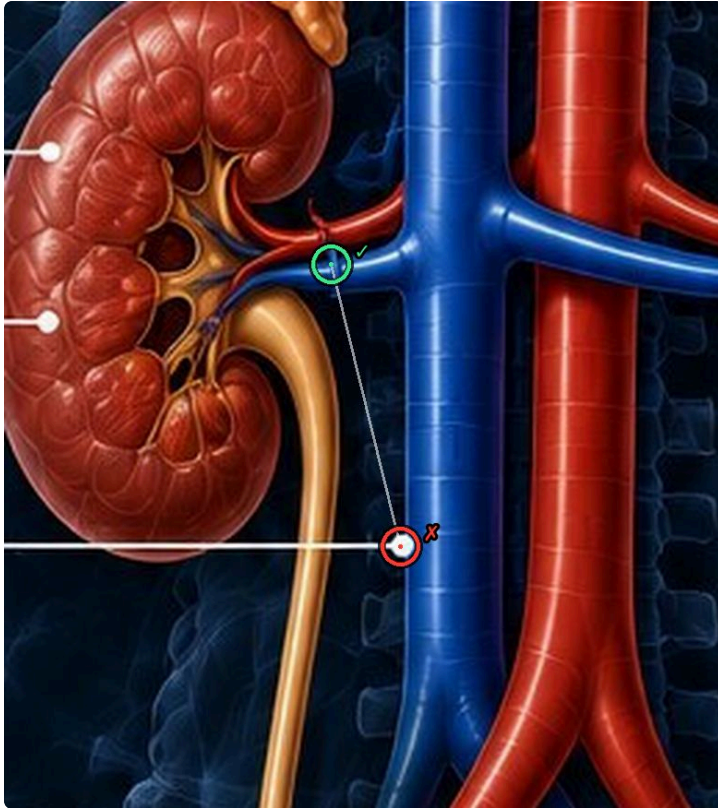
nephrolithiasis p2 — Renal artery

☒ ERASE (198,489)

kidney parenchyma, 59 px below the "Kidney" dot

☒ DRAW AT (295,458)

the red renal artery running from aorta to hilum



66

nephrolithiasis p2 — Renal vein

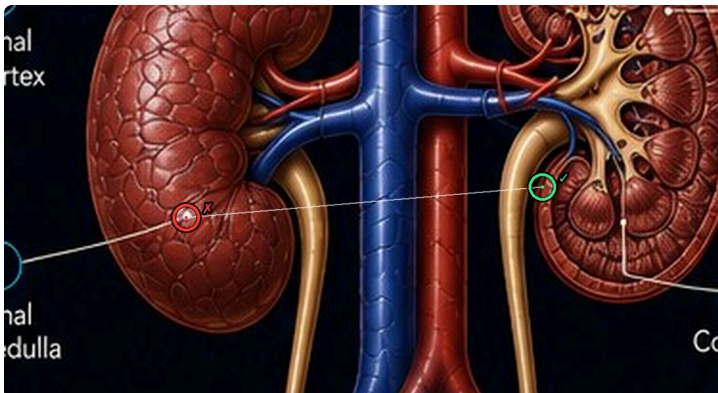
✗ ERASE (325,572)

the vertical blue IVC

✓ DRAW AT (300,470)

the horizontal renal vein joining the IVC at hilum level

| 102 px straight up the IVC.



67

hyperkalemia p2 — ② Renal Medulla

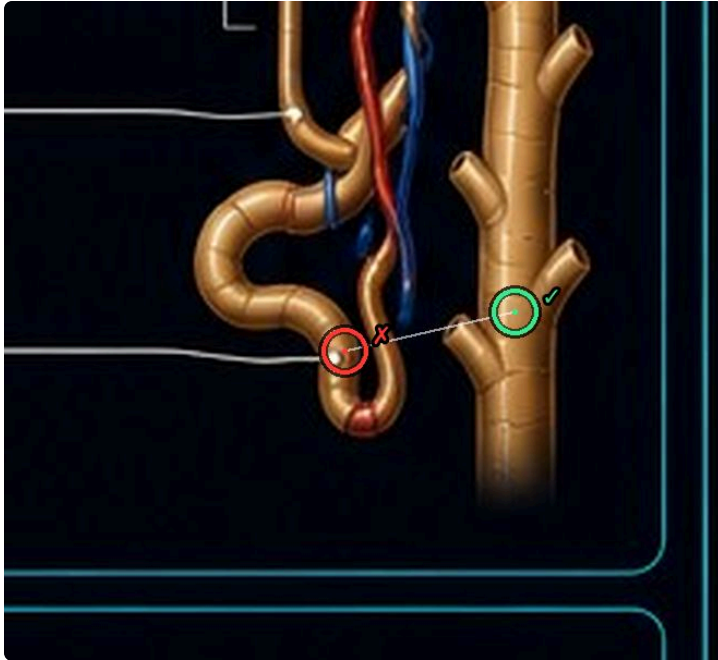
✗ ERASE (347,451)

the intact kidney's outer surface — capsule/cortex, the same tissue ① marks

✓ DRAW AT (533,435)

a striated pyramid in the CUT kidney opposite

| The intact kidney has no medulla showing anywhere on it.



68

hyperkalemia p2 — ④ Collecting Duct

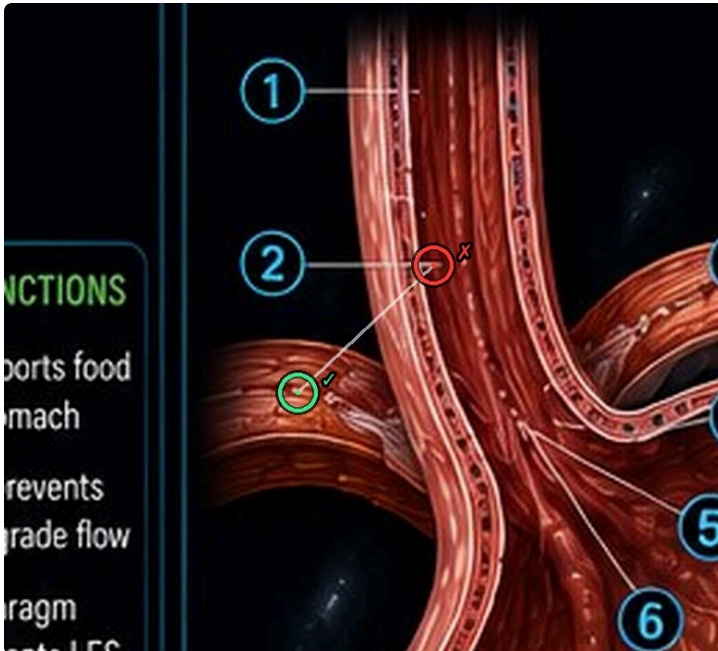
✗ ERASE (908,632)

the lower bend of the distal convoluted tubule coil

✓ DRAW AT (960,620)

the thick branched collecting duct

Three tan tubes at every level: loop limbs at x 888-894 and 909-913, collecting duct at 952-970.



69

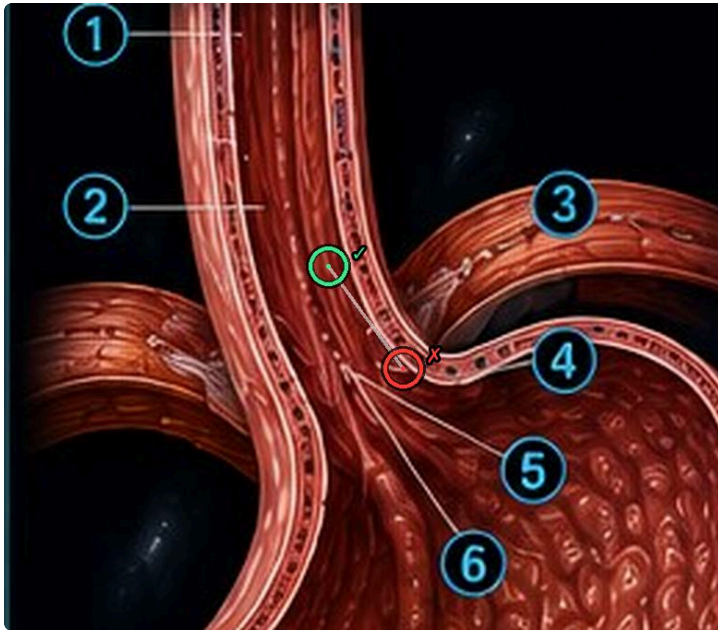
gerd p2 — ② Diaphragmatic crus (right)

✗ ERASE (444,319)

the oesophageal wall — the same wall ① marks 62 px above

✓ DRAW AT (395,365)

the left-hand crural muscle band, which is drawn and unlabelled



70

gerd p2 — ④ Lower esophageal sphincter

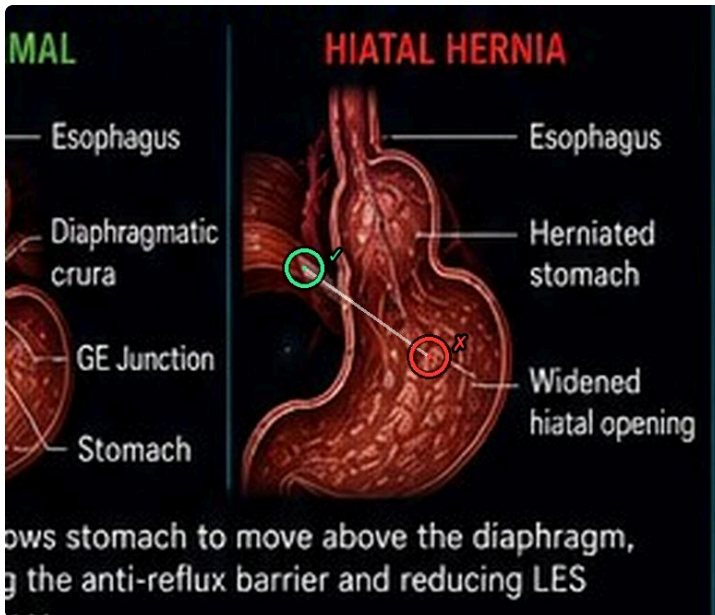
✗ ERASE (497,377)

a single point shared with ⑤ GEJ and ⑥ Angle of His — three markers, one spot

✓ DRAW AT (470,340)

the thickened distal oesophageal segment above the junction

⑤ and ⑥ can reasonably share the notch; the LES is a 3–4 cm segment and belongs above it.



71

gerd p2 — Widened hiatal opening

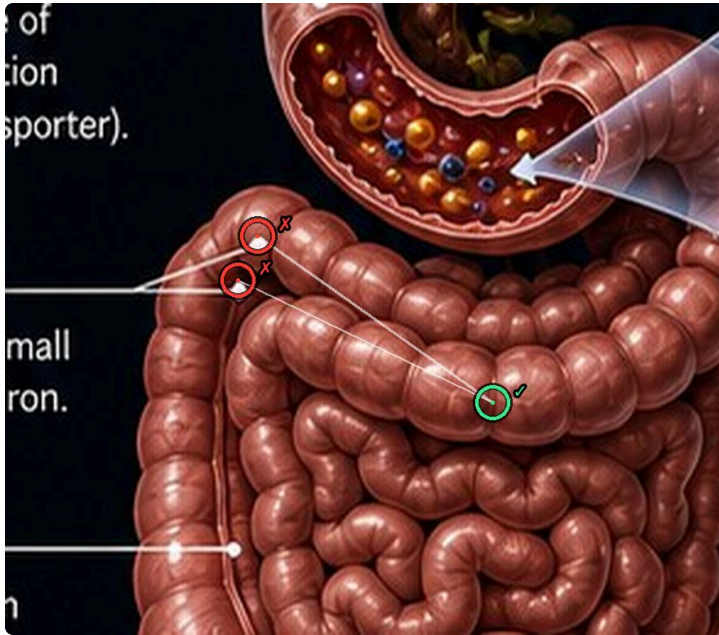
✗ ERASE (240,1032)

the mid gastric body of the herniated stomach

✓ DRAW AT (195,1000)

the gap between the crura at the neck of the hernia

48 px — more than a third of the panel's width.



72

iron-anemia p2 — JEJUNUM

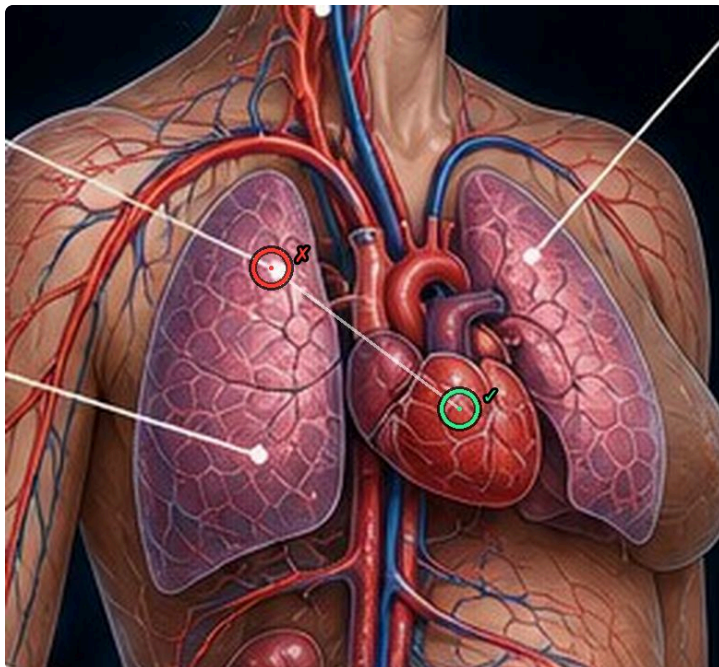
✗ ERASE (196,670) and (204,652)

TWO dots, both on the haustrated ascending colon, one on the taenia

✓ DRAW AT (300,720)

one dot on the smooth coiled small-bowel loops

The large bowel is deeply sacculated; the small bowel is smooth and narrower. ILEUM at (194,780) has the same problem.



73

preeclampsia p2 — HEART

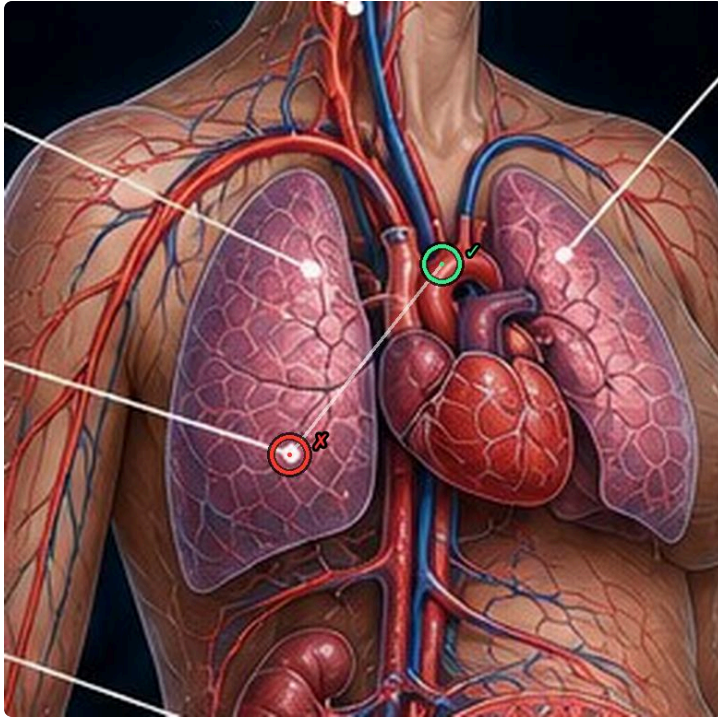
✗ ERASE (322,529)

the mid field of the patient's RIGHT LUNG

✓ DRAW AT (390,580)

the red ventricular mass

The heart and aorta are both drawn large and centrally and neither is labelled.



74

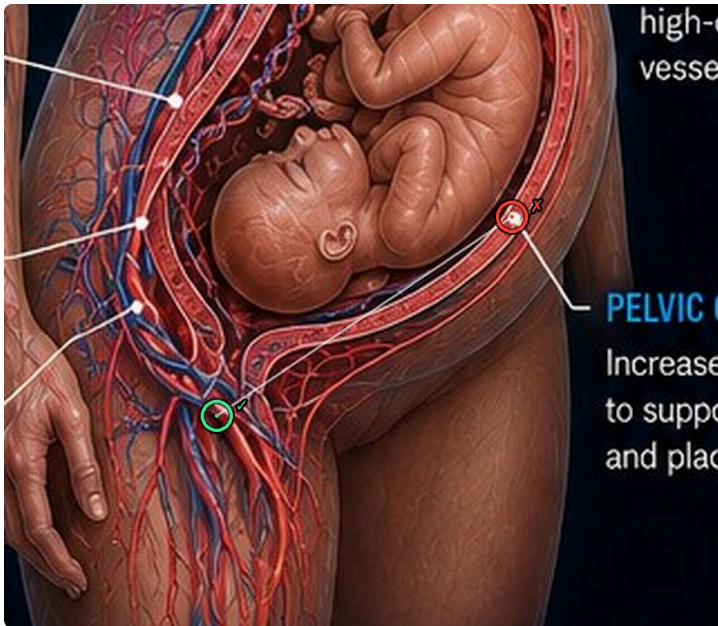
preeclampsia p2 — AORTA

✗ ERASE (317,597)

the lower field of the same lung, 68 px below the HEART dot

✓ DRAW AT (372,528)

the aortic arch



75

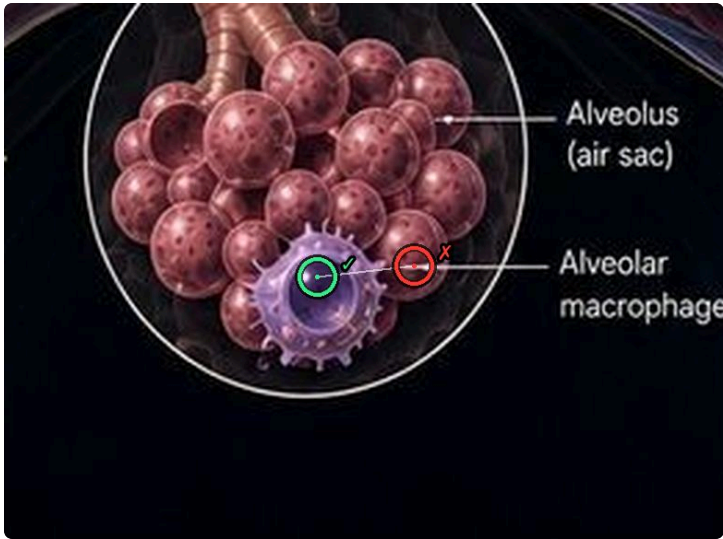
preeclampsia p2 — PELVIC CIRCULATION

✗ ERASE (452,841)

the uterine wall where it meets the fetal buttock — no maternal vessel within 80 px

✓ DRAW AT (320,930)

the maternal iliac vessel network



76

tb p2 — Alveolar macrophage

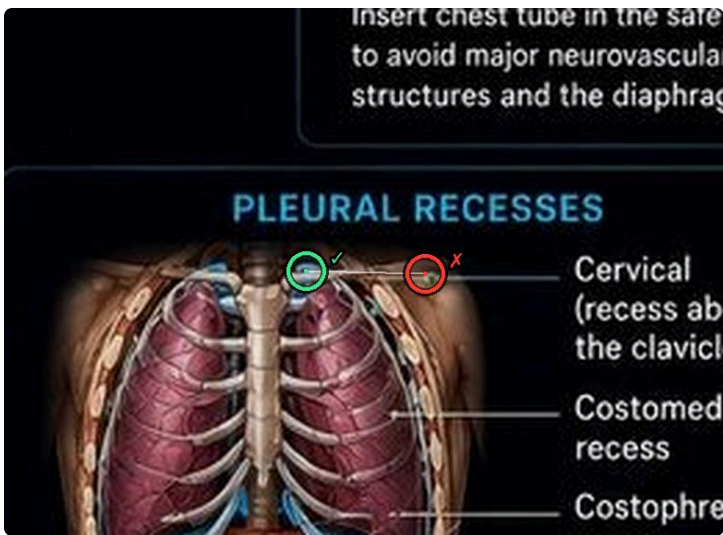
✗ ERASE (403,674)

a pink alveolar sphere — the structure “Alveolus (air sac)” names

✓ DRAW AT (368,678)

the purple spiculated macrophage

800×1200 page. The macrophage is the only purple object in the inset and 45 px across.



77

pneumothorax p2 — Cervical recess

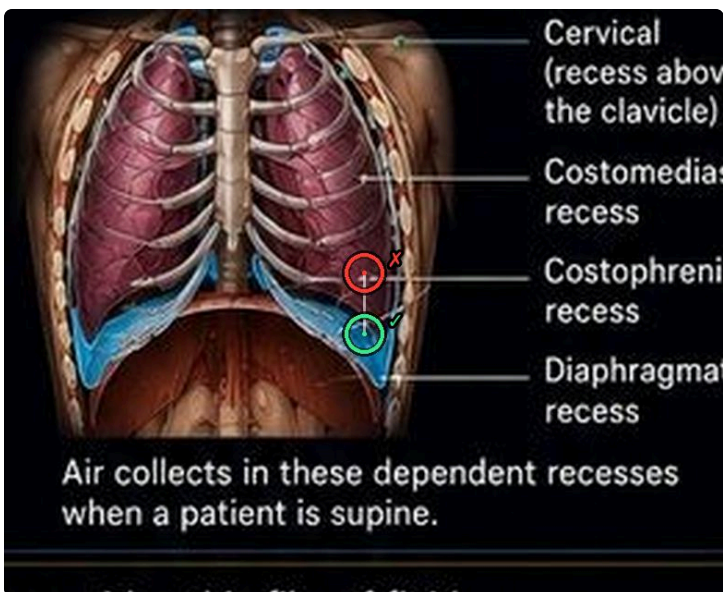
✗ ERASE (608,878)

shoulder soft tissue / first rib, lateral to the blue patch

✓ DRAW AT (565,877)

the blue apical recess above the clavicle

800×1200 page.



78

pneumothorax p2 — Costophrenic recess

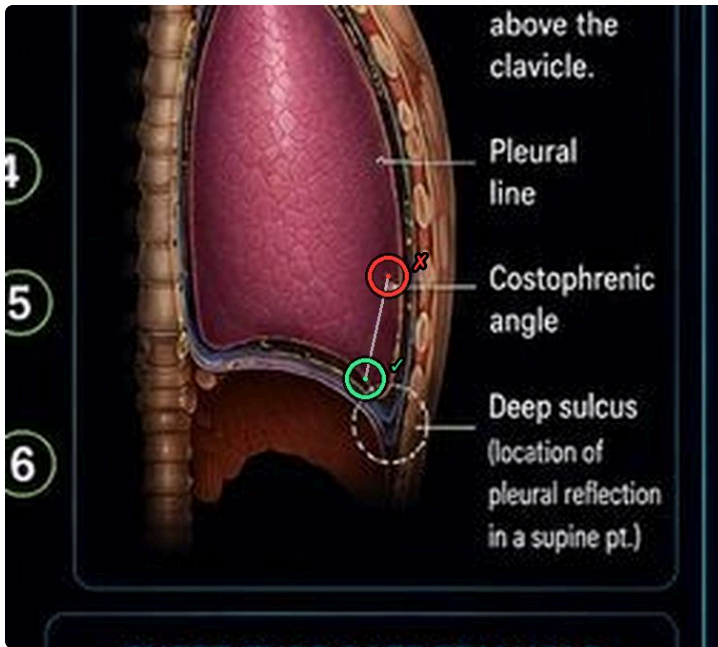
✗ ERASE (597,963)

lung parenchyma above the blue crescent

✓ DRAW AT (597,985)

the blue crescent at the lung base

“Diaphragmatic recess” at (605,1000) is correct and shows the form.



79

pneumothorax p2 — Costophrenic angle

☒ **ERASE** (682,402)

the lateral lung border, mid-zone

☒ **DRAW AT** (674,438)

the angle where the lung base meets the chest wall



80

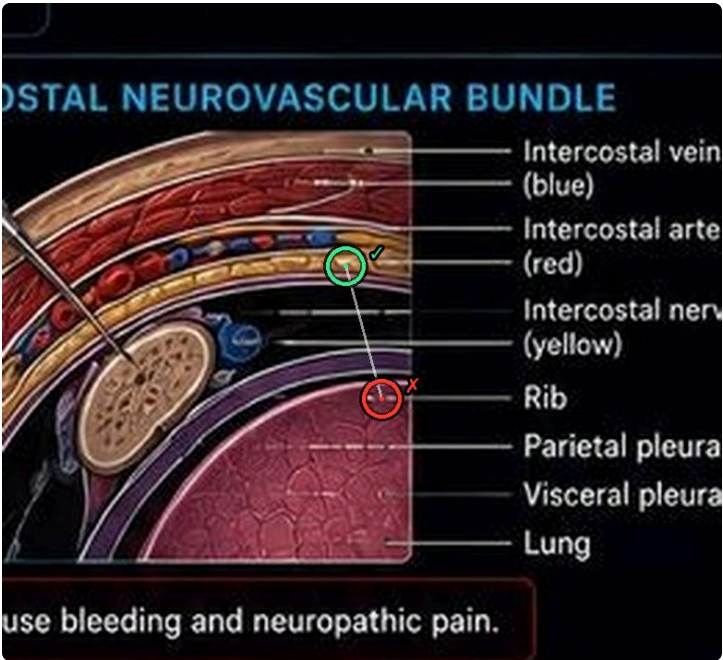
pneumothorax p2 — Intercostal vein (blue)

☒ **ERASE** (283,873)

the tan subcutaneous/periosteal band, 30 px above the groove

☒ **DRAW AT** (270,906)

the blue vein segment in the subcostal groove (x 258-283, y 902-912)



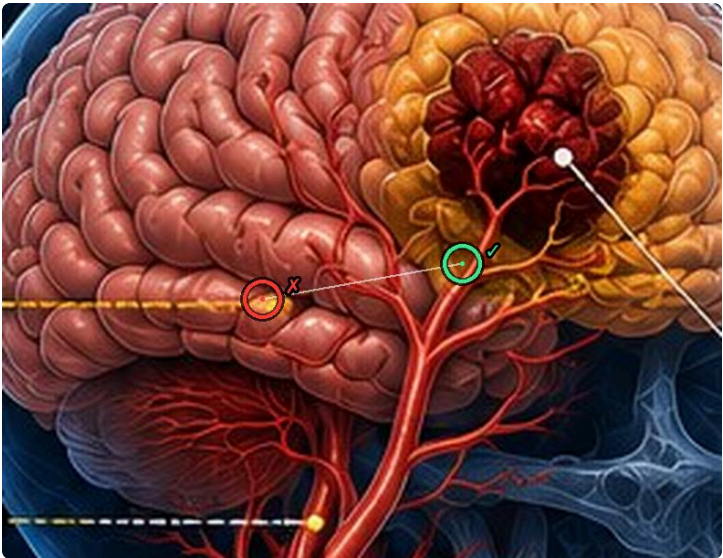
81

pneumothorax p2 — Rib (bundle panel)

ERASE (288,963)
magenta lung, 4 px inside the visceral pleura line

DRAW AT (275,915)
the gold rib band (y 912-919)

“Parietal pleura” and “Visceral pleura” on the same panel are also inside the lung — see the notes.



82

stroke p1 — Middle Cerebral Artery

ERASE (447,423)
a sulcus of plain pink, NON-infarcted cortex

DRAW AT (520,410)
the red trunk ascending into the ischemic zone

ALSO recolour this terminator WHITE. It is currently gold — the same amber the figure uses for the penumbra — which is why it reads as artwork.

C Group C — longer re-routes

The leader travels far enough that you are redrawing a stroke, not just a dot. Doable, but slower, and the erase crosses more artwork. Consider handing these to production.

PAGE	LABEL	FROM	TO	WHY IT IS HARDER
parkinsons p2	Brainstem	(245,726)	(358,600)	170 px. Old dot is on black — erase is trivial — but the whole leader must be redrawn.
parkinsons p2	Cerebellum	(303,760)	(455,595)	225 px, same shape.
stroke p1	Middle Cerebral Artery	(447,423)	(520,410)	Also recolour the terminator white — it is currently gold, the same amber the figure uses for the penumbra, which is why it reads as artwork.
cdiff p2	Superior mesenteric a.	(122,1173)	(192,1150)	70 px across the arcade.

PAGE	LABEL	FROM	TO	WHY IT IS HARDER
cdiff p2	Superior mesenteric v.	(478,1107)	(544,1120)	66 px.
cdiff p2	Inferior mesenteric v.	two dots, (481,1143) and (481,1160)	(510,1202)	Two terminators to erase; one to draw.
cdiff p2	Parasympathetic (vagus n.)	(851,1110)	(690,1140)	160 px across the whole colon frame.
cdiff p2	Portal vein (to liver)	(531,1202)	(545,1088)	Erase is easy — black ground — but the new route runs up the panel.
cdiff p2	Right colic a.	two tips, (122,1117) and (142,1118)	one tip on an arcade branch	Remove the orphan dash.
ckd p2	Afferent / Efferent arteriole	(865,1024) and (872,1063)	(775,970) and (770,1012)	Both leaders cross the whole corpuscle to reach the vascular pole on the far side.
seizure p2	Cortical networks	starts (531,891), arrowhead at (583,891) pointing outward	(555,845), arrowhead inward	Reverse the arrow. Not a move — the head is at the wrong end.
seizure p2	Basal ganglia	(496,452)	(552,452) or (545,478)	Crosses the magenta thalamus.
migraine p2	A — Dura mater	(521.5,306.5)	(470,283)	The dural band is only 6 px thick (y 280–285 at x 470). Precision job.
suicide p2	Raphe / others	—	—	See the per-page sheet; several leaders share one region.
hypothyroid p2	Capillary	(170,712)	(60,715)	110 px across the follicle. 800×1200 page.

D

Group D — do not attempt in Photoshop

The label names something the page does not draw

Nine rows. There is nothing to point at, so no leader move fixes them. These need an illustrator or a decision to drop the label. Full detail in `PRODUCTION-artwork-needed.md`.

PAGE	LABEL	WHAT IS MISSING	SUGGESTED
stroke p1	Circle of Willis, Basilar, PCA, ACA	all four — a lateral view cannot show basal vessels	add a basal inset, or drop the four
uti p2	Ⓢ Ureterovesical junction	no oblique intramural segment	draw it — the page's own relationships strip shows how
parkinsons p2	Substantia nigra pars reticulata	no SN band in the sagittal panel	copy from the page's own midbrain panel
parkinsons p2	Internal capsule	no white-matter lamina	draw one, or move the label to the coronal panel
schizophrenia p2	Pituitary Gland	no gland or stalk	draw it; the connector arrow is fine as is
migraine p2	D — Superficial temporal artery	no extracranial vessel anywhere	draw the STA in the scalp, or drop the label
hepatitis p2	Coronary ligament	a posterior structure in an anterior view	drop the label — it cannot be drawn here
hypothyroid p2	Hyooid bone	no bone above the thyroid cartilage	drop the label
dementia p2	Fornix	not differentiated from the callosal band	draw it, or fold into "Corpus callosum (fornix beneath)"

A legend entry whose marker was never drawn

Five. You *could* add a numbered badge in Photoshop — it is a circle and a digit on flat artwork — but check first whether the number should exist at all.

PAGE	MISSING MARKER
gerd p2	⑦ Stomach (cardia) — the artwork has ① to ⑥ only
meningitis p2	④ Cerebral aqueduct — no ④ anywhere
meningitis p2	⑥ Median aperture — no ⑥ anywhere
lymphoma p2	⑤ Mediastinal Lymph Nodes — the column runs 2,3,4,6,7,8,9
gout p2	⑤ Other metatarsals — no ⑤ badge on the foot

Blocked on a re-export, not on Photoshop

pneumothorax p2, rows 5-8. The four pleural layers on the main figure are **four pale lines one or two pixels wide with three-pixel gaps** — a terminator on that page is 4 px across, wider than the layer it names. They cannot be verified or drawn unambiguously at 800×1200. Re-export at 1024×1536 first. Same cause blocks four rows on *hypothyroid* p2 and one on *tb* p2.

Waiting on your read, not on anyone's tools

gdm p2, the PLACENTA (ENLARGED) panel. Either the umbilical vessel colours are inverted or the two leaders are on the wrong vessels — opposite fixes, and a render cannot distinguish them. Remember the umbilical circulation reverses the usual convention: the **vein** carries oxygenated blood and is drawn **red**.

The placenta on both obstetric pages. On *preeclampsia* p2 and *gdm* p2 the PLACENTA label lands on the uterine wall and I could not find a distinct placental disc on either. If it is not drawn, both are artwork rows.

osteoporosis p2, the lateral skeleton. Both legs overlap and I could not be certain which shaft belongs to the near leg. Confirm which bone is which before swapping Tibia and Fibula, and note that on the *anterior* skeleton both dots are on the slender bone — so that half needs a leader move as well as a text swap.

Compiled from 47 per-page production sheets, 11 August 2026. Each page's full reasoning, colour samples and unverified rows are in galleries-staging/PRODUCTION-<id>-p2.md on the claudes/anatomy-label-corrections-j58lf1 branch. 54 of the 81 pages in the work order are measured; 27 are not yet, so this list will grow.